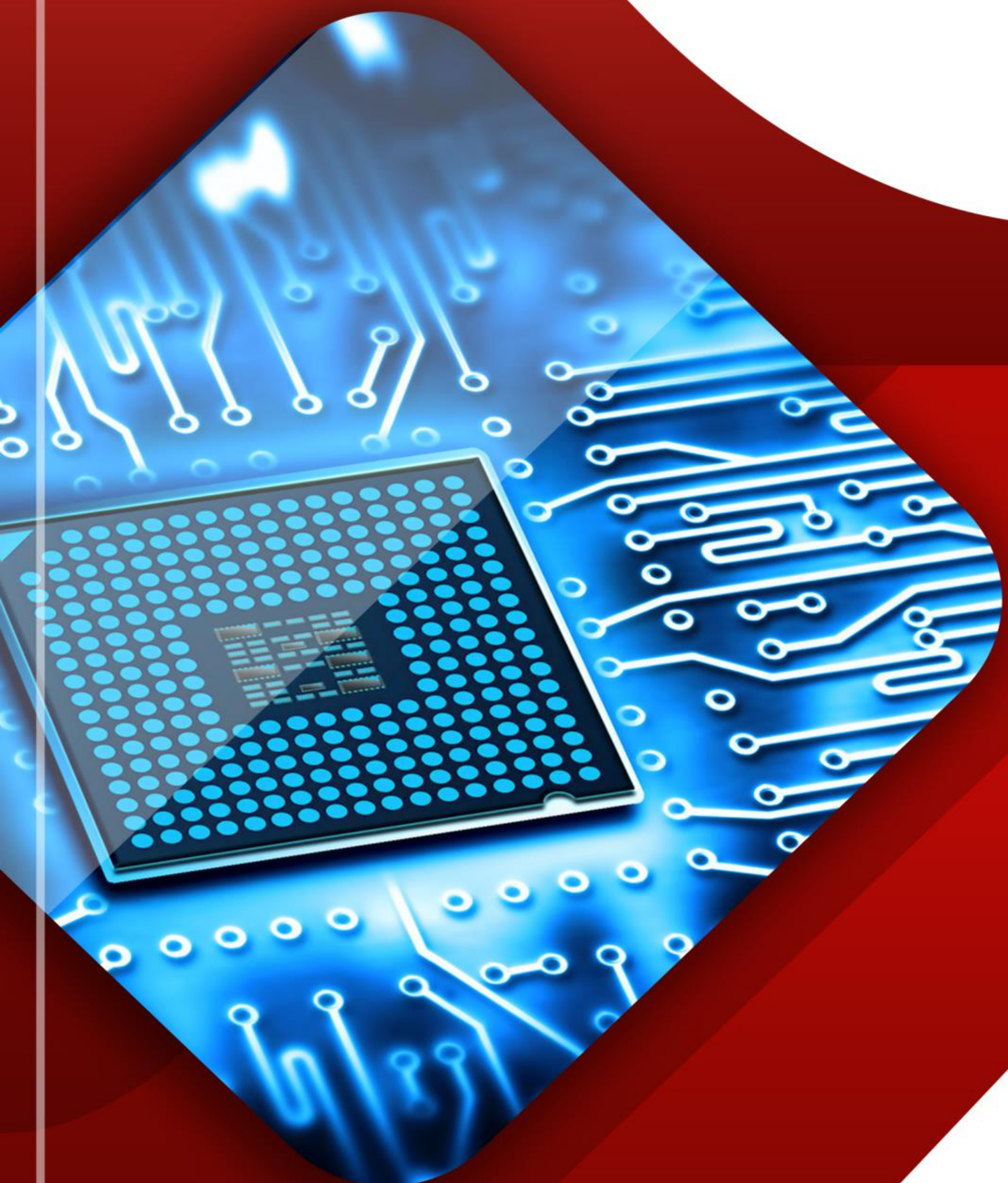
LABORATORY MANUAL IN

ELECTRONICS 3:

ELECTRONIC SYSTEM AND DESIGN



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SAFETY

Safety in the electrical laboratory, as everywhere else, is a matter of the knowledge of potential hazards, following safety precautions, and common sense. Observing safety precautions is important due to pronounced hazards in any electrical/computer engineering laboratory. Death is usually certain when 0.1 ampere or more flows through the head or upper thorax and have been fatal to persons with coronary conditions. The current depends on body resistance, the resistance between body and ground, and the voltage source. If the skin is wet, the heart is weak, the body contact with ground is large and direct, then 40 volts could be fatal. Therefore, never take a chance on "low" voltage. When working in a laboratory, injuries such as burns, broken bones, sprains, or damage to eyes are possible and precautions must be taken to avoid these as well as the much less common fatal electrical shock. Make sure that you have handy emergency phone numbers to call for assistance if necessary. If any safety questions arise, consult the lab demonstrator or technical assistant/technician for guidance and instructions. Observing proper safety precautions is important when working in the laboratory to prevent harm to yourself or others. The most common hazard is the electric shock which can be fatal if one is not careful.

Acquaint yourself with the location of the following safety items within the lab.

a. fire extinguisher
b. first aid kit
c. telephone and emergency numbers

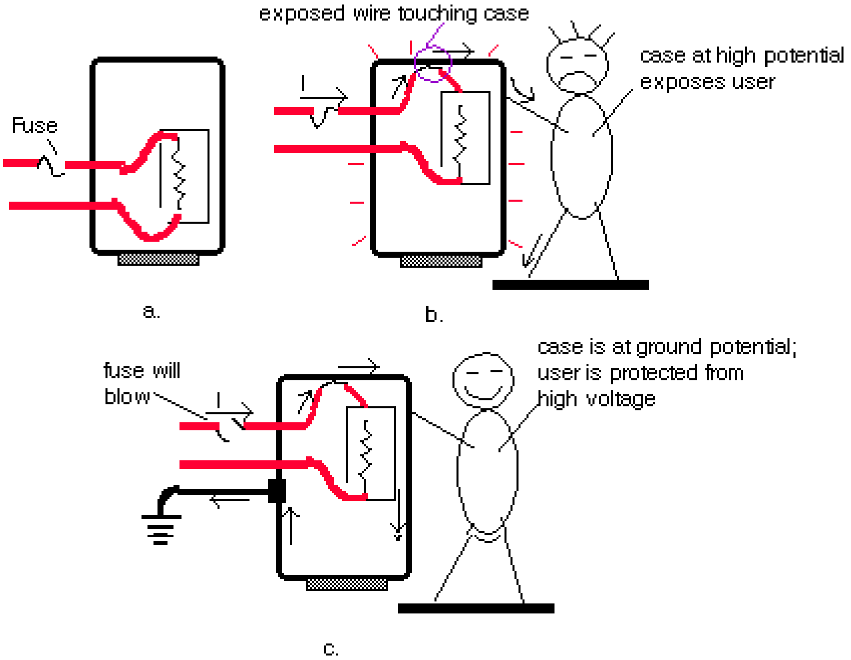
Electric shock

Shock is caused by passing an electric current through the human body. The severity depends mainly on the amount of current and is less function of the applied voltage. The threshold of electric shock is about 1 mA which usually gives an unpleasant tingling. For currents above 10 mA, severe muscle pain occurs and the victim can't let go of the conductor due to muscle spasm. Current between 100 mA and 200 mA (50 Hz AC) causes ventricular fibrillation of the heart and is most likely to be lethal.

What is the voltage required for a fatal current to flow? This depends on the skin resistance. Wet skin can have a resistance as low as 150 Ohm and dry skin may have a resistance of 15 kOhm. Arms and legs have a resistance of about 100 Ohm and the trunk 200 Ohm. This implies that 240 V can cause about 500 mA to flow in the body if the skin is wet and thus be fatal. In addition skin resistance falls quickly at the point of contact, so it is important to break the contact as quickly as possible to prevent the current from rising to lethal levels.

Equipment grounding

Grounding is very important. Improper grounding can be the source of errors, noise and a lot of trouble. Here we will focus on equipment grounding as a protection against electrical shocks. Electric instruments and appliances have equipments casings that are electrically insulated from the wires that carry the power. The isolation is provided by the insulation of the wires as shown in the figure a below. However, if the wire insulation gets damaged and makes contact to the casing, the casing will be at the high voltage supplied by the wires. If the user touches the instrument he or she will feel the high voltage. If, while standing on a wet floor, a user simultaneously comes in contact with the instrument case and a pipe or faucet connected to ground, a sizable current can flow through him or her, as shown in Figure b. However, if the case is connected to the ground by use of a third (ground) wire, the current will flow from the hot wire directly to the ground and bypass the user as illustrated in figure c.



Equipments with a three wire cord is thus much safer to use. The ground wire (3rd wire) which is connected to metal case is also connected to the earth ground (usually a pipe or bar in the ground) through the wall plug outlet.

Always observe the following safety precautions when working in the laboratory:

1. Do not work alone while working with high voltages or on energized electrical equipment or electrically operated machinery like a drill.

2. Power must be switched off whenever an experiment or project is being assembled, disassembled, or modified. **Discharge any high voltage points to grounds with a well insulated jumper. Remember that capacitors can store dangerous quantities of energy.**

3. Make measurements on live circuits or discharge capacitors with well insulated probes keeping one hand behind your back or in your pocket. Do not allow any part of your body to contact any part of the circuit or equipment connected to the circuit.

4. After switching power off, discharge any capacitors that were in the circuit. Do not trust supposedly discharged capacitors. Certain types of capacitors can build up a residual charge after being discharged. Use a shorting bar across the capacitor, and keep it connected until ready for use. If you use electrolytic capacitors, do not :
    - put excessive voltage across them

- put ac across them
- connect them in reverse polarity

5. Take extreme care when using tools that can cause short circuits if accidental contact is made to other circuit elements. Only tools with insulated handles should be used.

6. If a person comes in contact with a high voltage, immediately shut off power. Do not attempt to remove a person in contact with a high voltage unless you are insulated from them. If the victim is not breathing, apply CPR immediately continuing until he/she is revived, and have someone dial emergency numbers for assistance.

7. Check wire current carrying capacity if you will be using high currents. Also make sure your leads are rated to withstand the voltages you are using. This includes instrument leads.

8. Avoid simultaneous touching of any metal chassis used as an enclosure for your circuits and any pipes in the laboratory that may make contact with the earth, such as a water pipe. Use a floating voltmeter to measure the voltage from ground to the chassis to see if a hazardous potential difference exists.

9. Make sure that the lab instruments are at ground potential by using the ground terminal supplied on the instrument. Never handle wet, damp, or ungrounded electrical equipment.

10. Never touch electrical equipment while standing on a damp or metal floor.

11. Wearing a ring or watch can be hazardous in an electrical lab since such items make good electrodes for the human body.

12. When using rotating machinery, place neckties or necklaces inside your shirt or, better yet, remove them.

13. Never open field circuits of D-C motors because the resulting dangerously high speeds may cause a "mechanical explosion".

14. Keep your eyes away from arcing points. High intensity arcs may seriously impair your vision or a shower of molten copper may cause permanent eye injury.

15. Never operate the black circuit breakers on the main and branch circuit panels.

16. **In an emergency all power in the laboratory can be switched off by depressing the large red button on the main breaker panel. Locate it. It is to be used for emergencies only.**

17. **Chairs and stools should be kept under benches when not in use. Sit upright on chairs or stools keeping the feet on the floor. Be alert for wet floors near the stools.**

18. Horseplay, running, or practical jokes must not occur in the laboratory.

19. **Never use water on an electrical fire. If possible switch power off, then use CO_2 or a dry type fire extinguisher. Locate extinguishers and read operating instructions before an emergency occurs.**

20. Never plunge for a falling part of a live circuit such as leads or measuring equipment.

21. Never touch even one wire of a circuit; it may be hot.

22. Avoid heat dissipating surfaces of high wattage resistors and loads because they can cause severe burns.

23. Keep clear of rotating machinery.

Precautionary Steps Before Starting an Experiment so as Not to Waste Time Allocated

a) Read materials related to experiment before hand as preparation for pre-lab quiz and experimental calculation.

b) Make sure that apparatus to be used are in good condition. Seek help from technicians or the lab demonstrator in charge should any problem arises.
   - Power supply is working properly i.e. I_{max} (maximum current) LED indicator is disable. Maximum current will retard the dial movement and eventually damage the equipment. Two factors that will light up the LED indicator are short circuit and insufficient supply of current by the equipment itself. To monitor and maintain a constant power supply, the equipment must be connected to circuit during voltage measurement. DMM are not to be used simultaneously with oscilloscope to avert wrong results.
   - Digital multimeter (DMM) with low battery indicated is not to be used. By proper connection, check fuses functionality (especially important for current measurement). Comprehend the use of DMM for various functions. Verify measurements obtained with theoretical values calculated as it is quite often where 2 decimal point reading and 3 decimal point reading are very much deviated.
   - The functionality of voltage waveform generators are to be understood. Make sure that frequency desired is displayed by selecting appropriate multiplier knob. Improper settings (ie selected knob is not set at minimum (in direction of CAL – calibrate) at the bottom of knob) might result in misleading values and hence incorrect results. Avoid connecting oscilloscope together with DMM as this will lead to erroneous result.
   - Make sure both analog and digital oscilloscopes are properly calibrated by positioning sweep variables for VOLT / DIV in direction of CAL. Calibration can also be achieved by stand alone operation where coaxial cable connects CH1 to bottom left hand terminal of oscilloscope. This procedure also verifies coaxial cable continuity.

c) Internal circuitry configuration of breadboard or Vero board should be at students' fingertips (i.e. holes are connected horizontally not vertically for the main part with engravings disconnecting in-line holes).

d) Students should be rest assured that measured values (theoretical values) of discrete components retrieved i.e. resistor, capacitor and inductor are in accordance the required ones.

e) Continuity check of connecter or wire using DMM should be performed prior to proceeding an experiment. Minimize wires usage to avert mistakes.

f) It is unethical and wrong for students to falsify results as to make them appear exactly consistent with theoretical calculations.

Safety Instructions for the Electronics 3

Agreement

I have read and agree to follow all of the safety rules. I realize that I must obey these rules to insure my own safety, and that of my fellow students, instructors and teaching assistants. I will cooperate to the fullest extent with my instructor and fellow students to maintain a safe lab environment. I will also closely follow the oral and written instructions provided by the instructor. I am aware that any violation of these rules that result in unsafe conduct in the laboratory or misbehavior on my part, may result in being removed from the laboratory, receiving a failing grade, and/or dismissal from the course.

Complete Name:____

Course : ____

Signature : ____

Date:____

Criteria	Unsatisfactory (1-3)	Satisfactory (4-6)	Good (7-9)	Excellent (10)
Completion (20%)	- less than 50% of the requirement has been completed	- between 50-65% of the requirement has been completed	- between 65-95% of the requirement has been completed	- between 100% of the requirement has been completed
Presentation and Organization (10%)	- no name, date, or requirement title included - poor use of materials - disorganized and messy	- name, date, requirement title included - use of materials that is easy to read - organized work	- name, date, requirement title included - good use of materials - organized work	- name, date, requirement title included - excellent use of materials - effective use of materials - creatively organized work
Correctness (30%)	- Circuit does not execute due to errors - no error checking code included - no testing has been completed	- Circuit executes without errors - Circuit handles some special cases - some testing has been completed	- Circuit executes without errors - Circuit handles most special cases - thorough testing has been completed	- Circuit executes without errors - Circuit handles all special cases - thorough and organized testing has been completed and output from test cases is included
Creativity (30%)	- Circuit uses a difficult and inefficient solution - designer has not considered alternate solutions	- Circuit uses a logical solution that is easy to follow but it is not the most efficient - designer has considered alternate solutions	- Circuit uses an efficient and easy to follow solution (i.e. no confusing tricks) - designer has considered alternate solution and has chosen the most efficient	- Circuit uses solution that is easy to understand and maintain - designer has analyzed many alternate solutions and has chosen the most efficient - designer has included the reasons for the solution chosen
Documentation (10%)	- no documentation included	- basic documentation has been completed	- program has been clearly documented	- clearly and effectively documented

ELECTRONICS 3

1. Basic Guidelines
2. Lab Instructions
3. Lab Reports
4. Grading______________
5. Schedule and Experiment No. (Title)

1. Basic Guidelines

All experiments in this manual have been tried and proven and should give you little trouble in normal laboratory circumstances. However, a few guidelines will help you conduct the experiments quickly and successfully.

1. Each experiment has been written so that you follow a structured logical sequence meant to lead you to a specific set of conclusions. Be sure to follow the procedural steps in the order which they are written.
2. Read the entire experiment and research any required theory beforehand. Many times an experiment takes longer that one class period simply because a student is not well prepared.
3. Once the circuit is connected, if it appears “dead’’ spend few moments checking for obvious faults. Some common simple errors are: power not applied, switch off, faulty components, lose connection, etc. Generally the problems are with the operator and not the equipment.
4. When making measurements, check for their sensibility.
5. **It’s unethical to “fiddle” or alter your results to make them appear exactly consistent with theoretical calculations.**

2. Lab Instructions

1. Each student group consists of a maximum of 5 students. Each group is responsible in submitting 1 lab report upon completion of each experiment.
2. Students are to wear proper attire i.e. shoe or sandal instead of slipper. Excessive jewelries are not advisable as they might cause electrical shock.
3. Personal belongings i.e. bags, etc are to be put at the racks provided. Student groups are required to wire up their circuits in accordance with the diagram given in each experiment.
4. A permanent record in ink of observations as well as results should be maintained by each student and enclosed with the report.
5. The **recorded data** and **observations** from the lab manual need to be **approved** and **signed** by the **lab instructor** upon completion of each experiment.
6. Before beginning connecting up, it is essential to check that all sources of supply at the bench are switched off.
7. Starts connecting up the experiment circuit by wiring up the main circuit path, then add the parallel branches as indicated in the circuit diagram.
8. After the circuit has been connected correctly, remove all unused leads from the experiment area, set the voltage supplies at the minimum value, and check the meters are set for the intended mode of operation.
9. The students may ask the lab instructor to check the correctness of their circuit before switching on.
10. When the experiment has been satisfactory completed and the results approved by the instructor, the students may disconnect the circuit and return the components and instruments to the counter tidily. Chairs are to be slid in properly.

3. Grading

The work in the Electronics related lab carries 40% of total marks for the subjec. The distribution of marks for Electronics Lab is as follows:

Midterm Exam Hands-On	25 %
Final Exam Hands -On	25 %
Lab Report	25%
Project (Simulation and Hardware)	25%
Total	**100%**

Table of Content

S. NO	NAME OF THE EXPERIMENT	MARKS	SIGNATURE
1	Study of characteristics of SCR, MOSFET & IGBT		
2	Single Phase Half controlled converter with R load		
3	Single Phase fully controlled bridge converter with R and RL loads		
4	Three Phase half controlled bridge converter with R load		
5	Single Phase AC Voltage Controller with R and RL Loads		
6	Single Phase Cycloconverters with R and RL loads		
7	Single Phase series inverter with R and RL loads		
8	DC Jones chopper with R and RL Loads		
9	Single-phase full converter using RLE loads and single phase AC voltage controller using RLE loads using PSPICE		
10	Resonant pulse commutation circuit and Buck chopper using PSPICE.		
11	Single phase Inverter with PWM control using PSPICE		
12	Single Phase dual converter with RL loads		
13	Single Phase Mc-Murray converter with R and RL loads		
14	Gate firing circuits for SCR's		
15	Single Phase Parallel inverter with R and RL loads		

Introduction

The experiments listed encompass a comprehensive study of power electronics and its applications, focusing on various semiconductor devices, converters, inverters, and their control methods. These experiments provide hands-on experience with crucial components and circuits used in modern power electronics, offering insights into the operation, characteristics, and applications of devices like SCRs, MOSFETs, IGBTs, and various types of converters and inverters. Utilizing simulation tools like PSPICE in some experiments further enhances understanding by allowing for detailed analysis and modeling of complex circuits.

Learning Objectives

1. Study of Characteristics of SCR, MOSFET & IGBT:
   Understand the operational characteristics and differences between SCRs, MOSFETs, and IGBTs.
   Analyze the current-voltage characteristics and switching behavior of these devices.

2. Single Phase Half Controlled Converter with R Load:
   Learn the principles and operation of single-phase half-controlled converters.
   Analyze the converter's performance with resistive loads and understand its control strategies.
   Single Phase Fully Controlled Bridge Converter with R and RL Loads:

3. Understand the operation and control of single-phase fully controlled bridge converters.
   Study the impact of resistive and inductive loads on converter performance.

4. Three Phase Half Controlled Bridge Converter with R Load:
   Gain insights into three-phase half-controlled converters and their advantages.
   Analyze the converter's behavior with resistive loads.

5. Single Phase AC Voltage Controller with R and RL Loads:

   Understand the operation of single-phase AC voltage controllers.
   Study the effects of resistive and inductive loads on the performance of AC voltage controllers.
6. Single Phase Cycloconverters with R and RL Loads:

   Learn about the principles and applications of cycloconverters.
   Analyze how cycloconverters perform with resistive and inductive loads.

7. Single Phase Series Inverter with R and RL Loads:
   Understand the operation of single-phase series inverters.
8. Study the inverter's performance and output characteristics with different loads.
   DC Jones Chopper with R and RL Loads:
   Explore the operation and applications of DC Jones choppers.
   Analyze the chopper performance with various load types.
9. Single-phase Full Converter and AC Voltage Controller Using PSPICE:
   Use PSPICE to simulate and analyze single-phase full converters and AC voltage controllers.
   Understand the impact of RLE loads on the operation of these circuits.
   Resonant Pulse Commutation Circuit and Buck Chopper Using PSPICE:

10. Study resonant pulse commutation circuits and buck choppers through PSPICE simulations.
    Analyze the operation and efficiency of these circuits.
    Single Phase Inverter with PWM Control Using PSPICE:

11. Understand the principles of PWM control in single-phase inverters.
    Use PSPICE to simulate and analyze the performance of PWM-controlled inverters.
    Single Phase Dual Converter with RL Loads:

12. Learn about the operation and control of single-phase dual converters.
    Analyze how these converters perform with inductive loads.
    Single Phase Mc-Murray Converter with R and RL Loads:

13. Study the Mc-Murray converter and its applications.
    Understand the converter's performance with different types of loads.
    Gate Firing Circuits for SCRs:

14. Understand the design and operation of gate firing circuits for SCRs.
    Learn about the techniques used for effective gate signal generation.

15. Single Phase Parallel Inverter with R and RL Loads:
    Explore the principles and operation of single-phase parallel inverters.
    Study the inverter's behavior and efficiency with various loads.

EXPERIMENT – 1(A)

STUDY OF CHARACTERISTICS OF SCR, MOSFET & IGBT

SCR CHARACTERISTICS

Introduction

The experiment on the "Study of Characteristics of SCR, MOSFET, and IGBT" is an essential part of the curriculum in electronics and electrical engineering, focusing on understanding the operational characteristics of key semiconductor devices used in power electronics. Silicon Controlled Rectifiers (SCRs), Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs), and Insulated Gate Bipolar Transistors (IGBTs) are fundamental components in a wide range of applications, from industrial motor control to power management in electronic devices.

This experiment involves examining the electrical characteristics of these devices, such as their voltage-current relationships, switching behavior, triggering requirements, and thermal properties. By conducting this experiment, students gain a deeper understanding of each device's unique features, operation principles, and practical applications, which is crucial for designing and troubleshooting electronic circuits and systems.

Learning Objectives

1. Understanding Device Fundamentals: Develop a comprehensive understanding of the construction, operational principles, and key characteristics of SCRs, MOSFETs, and IGBTs.

2. Characteristic Curve Analysis: Learn to measure and analyze the VI (Voltage-Current) characteristics of each device, understanding the significance of parameters like breakdown voltage, on-state current, and leakage current.

3. Switching Behavior and Control: Study the switching behavior of these devices, focusing on how they are turned on and off, and the importance of gate or base control in their operation.

4. Triggering Mechanisms: Understand the different triggering mechanisms for SCRs and the gate control for MOSFETs and IGBTs, including the required voltage and current levels.

5. Comparative Analysis: Compare SCRs, MOSFETs, and IGBTs in terms of their operational efficiencies, switching speeds, power handling capabilities, and suitability for various applications.

6. Thermal Characteristics: Explore the thermal characteristics and heat management requirements of these devices, understanding the importance of thermal stability in power electronics.

7. Safety in Handling Semiconductor Devices: Emphasize the importance of safety and proper handling procedures when working with semiconductor devices, especially under high voltage and current conditions.

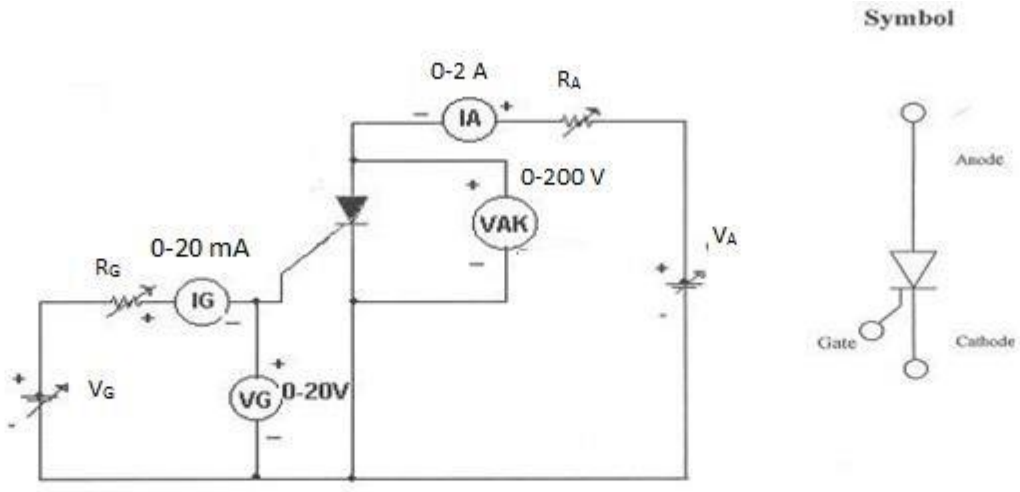
8. Real-World Applications: Recognize the practical applications of SCRs, MOSFETs, and IGBTs in various domains, such as in power supplies, motor controls, and renewable energy systems.

9. Problem-Solving and Critical Thinking: Enhance problem-solving skills by diagnosing issues in circuits involving these devices and proposing solutions based on their characteristics.
10. To plot the V - I Characteristics of SCR, MOSFET & IGBT.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	SCR characteristics Trainer	-	-	1
2	Patch chords	-	-	
3	DC Voltmeter	-	Digital	1
4	DC Ammeter	-	Digital	1

CIRCUIT DIAGRAM:



Study of Characteristics of SCR

PROCEDURE:

V - I CHARACTERISTICS:

1. Make all connections as per the circuit diagram.
2. Initially keep V_G & V_A at minimum position and R_1 & R_2 maximum position.
3. Adjust Gate current I_g to some constant by varying the V_G or R_G.
4. Now slowly vary $V_A$and observe Anode to Cathode voltage V_{AK} and Anode current I_A.
5. Tabulate the readings of Anode to Cathode voltage V_{AK} and Anode current I_A.
6. Repeat the above procedure for different Gate current I_g.

GATE TRIGGRING AND FINDING V_G AND I_G:-

1. Keep all positions at minimum.
2. Set Anode to Cathode voltage V_{AK} to some volts say 15V.

3. Now slowly vary the V_G voltage till the SCR triggers and note down the reading of gate current(I_G) and Gate Cathode voltage(V_{GK}) and rise of anode current I_A.
4. Repeat the same for different Anode to Cathode voltage and find VAK and I_G values.

TO FIND LATCHING CURRENT:

1. Keep R_2 at middle position.
2. Apply 20V to the Anode to cathode by varying V_2.
3. Rise the Vg voltage by varying V_G till the device turns ON indicated by sudden rise in I_A.
   At what current SCR trigger it is the minimum gate current required to turn ON the SCR.
4. Now set R_A at maximum position, then SCR turns OFF, if it is not turned off reduce V_A up to turn off the device and put the gate voltage.
5. Now decrease the R_A slowly, to increase the Anode current gradually in steps.
6. At each and every step, put OFF and ON the gate voltage switches V_G. If the Anode current is greater than the latching current of the device, the device says ON even after switch OFF S1, otherwise device goes to blocking mode as soon as the gate switch is put OFF.
7. If $I_A > I_L$ then, the device remains in ON state and note that anode current as latching current.
8. Take small steps to get accurate latching current value.

TO FIND HOLDING CURRENT:

1. Now increase load current from latching current level by varying R_A & V_A.
2. Switch OFF the gate voltage switch S1 permanently (now the device is in ON state).
3. Now increase load resistance(R2), so that anode current reducing, at some anode current the device goes to turn off .Note that anode current as holding current.
4. Take small steps to get accurate holding current value.
5. Observe that $I_H < I_L$.

TABULAR COLUMN:

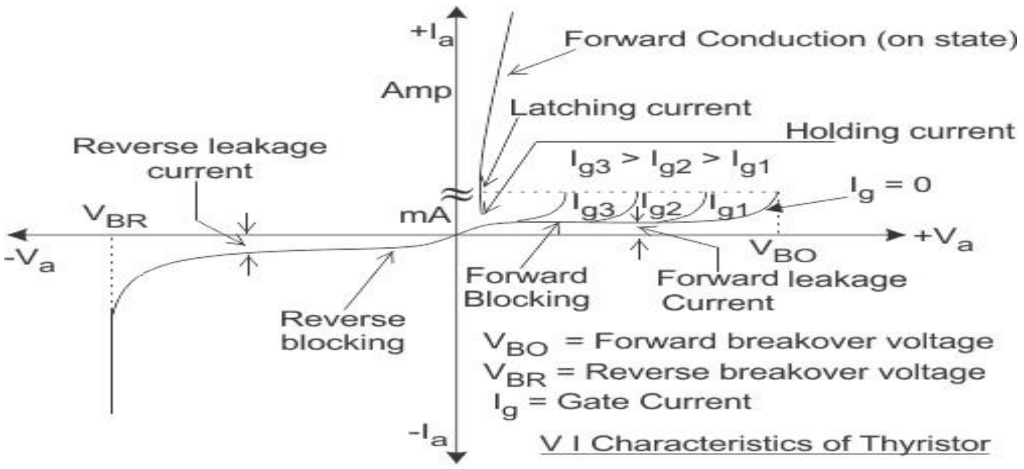
S. No	I_G= Amps	
	VAK (Volts)	I_A (Amps)
1		
2		
3		
4		
5		

S. No	I_G= Amps	
	V_{AK} (Volts)	I_A (Amps)
1		
2		
3		
4		
5		

S. No	V_{AK} = (Volts)	
	V_{GK} = (Volts)	I_G= (Amps)
1		
2		
3		
4		
5		

S. No	V_{AK} = (Volts)	
	V_{GK} = Volts	I_G= Amps
1		
2		
3		
4		
5		

MODEL GRAPH:



V- I Characteristics of SCR

PRE LAB QUESTIONS:

1. What is a Thyristor? Draw the structure of an SCR?
2. What are the different methods of turning on an SCR?
3. What is Forward break over voltage? Reverse break over voltage?
4. What are modes of working of an SCR?
5. Draw the V-I characteristics of SCR.
6. Differentiate between holding and latching currents.
7. Why is dv /dt technique not used for triggering?

POST LAB QUESTIONS:

1. Why is V_{bo} greater than V_{br}?
2. Why does high power dissipation occur in reverse blocking mode?
3. Why shouldn't positive gate signal be applied during reverse blocking Mode?
4. Explain reverse current $I_{rev.}$.
5. What happens when gate drive is applied?
6. Why should the gate signal be removed after turn on?
7. Is a gate signal required when reverse biased?
8. What are applications of SCR?

EXPERIMENT – 1(B)

MOSFET CHARACTERISTICS

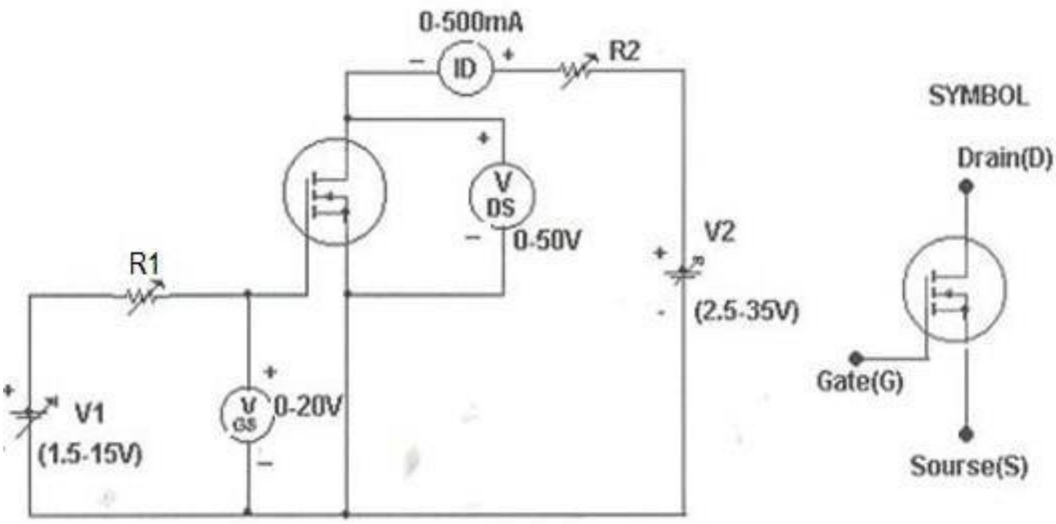
AIM:

To study the output and transfer characteristics of MOSFET

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	MOSFET characteristics Trainer			
2	Patch chords			
3	DC Voltmeter			
4	DC Ammeter			

CIRCUIT DIAGRAM:



Study of Characteristics of MOSFET

PROCEDURE:

TRANSFER CHARACTERISTICS:

1. Make all connections as per the circuit diagram.
2. Initially keep V_1 & V_2 at minimum position and R_1 & R_2 middle position.
3. Set V_{DS} to some say 10V.
4. Slowly vary Gate source voltage V_{GS} by varying V1.
5. Note down I_D and V_{GS} readings for each step.

6. Repeat above procedure for 20V & 30V of V_{DS}. Draw Graph between I_D & V_{GS}.

OUTPUT CHARACTERISTICS:

1. Initially set V_{GS} to some value say 3V by varying V1.
2. Slowly vary V2 and note down I_D and V_{DS}.
3. At particular value of V_{GS} there a pinch off voltage between drain and source.
   If $V_{DS} < V_P$ device works in the constant resistance region and I_O is directly proportional to V_{DS}. If $V_{DS} > V_P$ device works in the constant current region.
4. Repeat above procedure for different values of V_{GS} and draw graph between I_D *vs* V_{DS}.

TABULAR COLUMN:

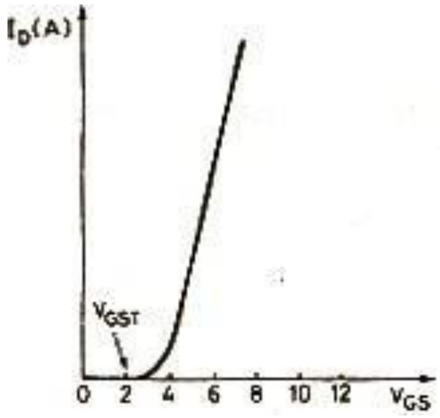
S. No.	V_{GS} = VOLTS	
	VDS (Volts)	I_D (Amps)
1		
2		
3		
4		
5		

S. No	V_{GS} = VOLTS	
	V_{DS} (Volts)	I_D (Amps)
1		
2		
3		
4		
5		

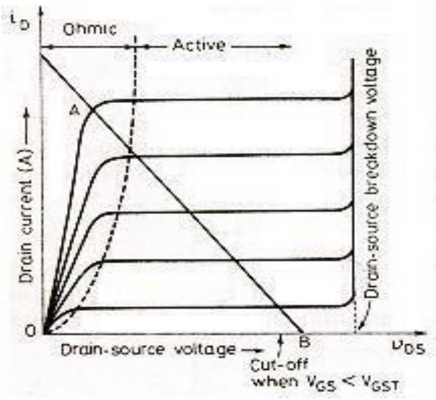
S. No	V_{DS}= (Volts)	
	V_{GS} (V)	I_D(A)
1		
2		
3		
4		
5		

S. No	V_{DS} = (Volts)	
	V_{GS} (V)	I_D(A)
1		
2		
3		
4		
5		

MODEL GRAPH:



Transfer Characteristic of MOSFET



Output Characteristics of MOSFET

PRE LAB QUESTIONS:

1. What are MOSFET's?
2. Draw the symbol of MOSFET.
3. What is the difference between MOSFET and BJT?
4. What is the difference between JFET and MOSFET?
5. Draw the structure of MOSFET.
6. What are the types of MOSFET?
7. What is the difference between depletion and enhancement MOSFET?

POST LAB QUESTIONS:

1. How does n - drift region affect MOSFET?
2. How MOSFET are suitable for low power high frequency applications?
3. What are the requirements of gate drive in MOSFET?
4. Draw the switching model of MOSFET.
5. What is rise time and fall time?
6. In which region does the MOSFET used as a switch?
7. Why are MOSFET's mainly used for low power applications?
8. How is MOSFET turned off?
9. What are the advantages of vertical structure of MOSFET?
10. What are the merits of MOSFET?

EXPERIMENT – 1(C)

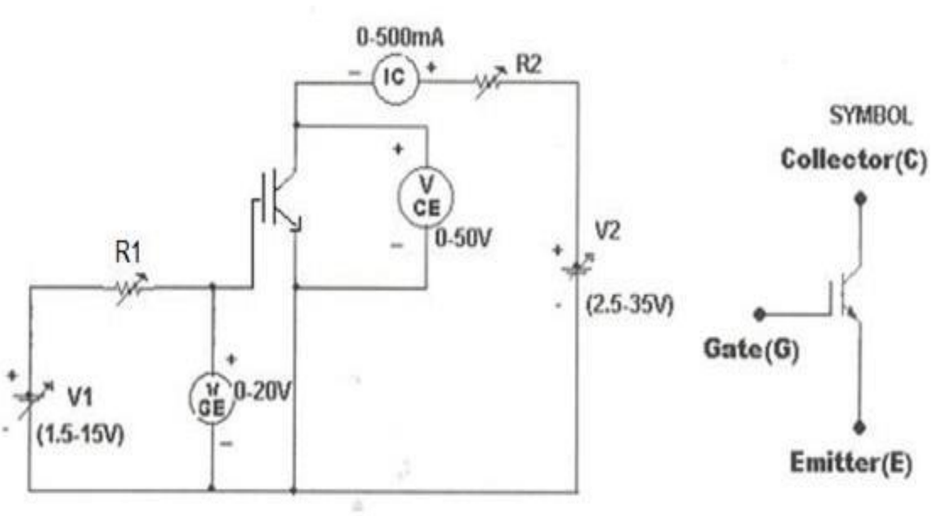
IGBT CHARACTERISTICS

AIM: To study the output and transfer characteristics of IGBT.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	IGBT characteristics Trainer			
2	Patch chords			
3	DC Voltmeter			
4	DC Ammeter			

CIRCUIT DIAGRAM:



Study of Characteristics IGBT

PROCEDURE:

TRANSFER CHARACTERISTICS:

1. Make all connections as per the circuit diagram.
2. Initially keep V_1 & V_2 at minimum position and R_1 & R_2 middle position.
3. Set V_{CE} to some say 10V.
4. Slowly vary Gate Emitter voltage V_{GE} by varying V1.
5. Note down I_C and V_{GE} readings for each step.
6. Repeat above procedure for 20V & 25V of $V_{DS.}$ Draw Graph between I_D & V_{GS}.

OUTPUT CHARACTERISTICS:

1. Initially set V_{GE} to some value say 5V by varying V1.
2. Slowly vary V2 and note down I_C and V_{CE} readings.
3. At particular value of V_{GS} there a pinch off voltage V_P between Collector and Emitter.

If $V_{CE} < V_P$ device works in the constant resistance region and I_C is directly proportional to V_{CE}. If $V_{CE} > V_P$ device works in the constant current region. 4. Repeat above procedure for different values of V_{GE} and draw graph between I_C *vs* V_{GE}.

TABULAR COLUMN:

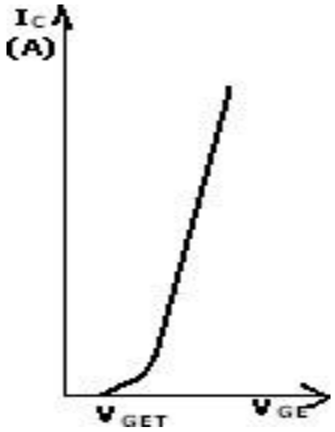
S. No	V_{CE} = Volts	
	VGE (Volts)	I_C (Amps)
1		
2		
3		
4		
5		

S. No	V_{CE} =Volts	
	VGE (Volts)	I_C (Amps)
1		
2		
3		
4		
5		

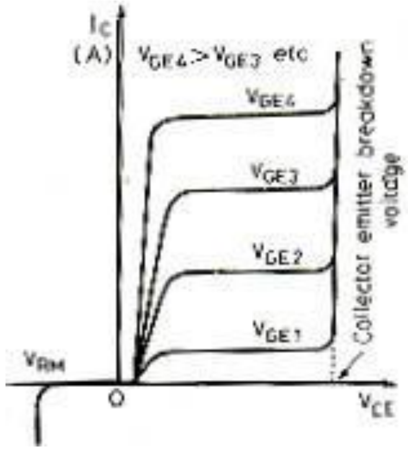
S. No	V_{GE}= (Volts)	
	V_{CE} (Volts)	I_C(Amps)
1		
2		
3		
4		
5		

S. No	V_{GE} = (Volts)	
	V_{CE} (Volts)	I_C(Amps)
1		
2		
3		
4		
5		

MODEL GRAPH:



Transfer Characteristics of IGBT



Output Characteristics of IGBT

PRE LAB QUESTIONS:

1. What is IGBT? What is the difference between an IGBT and SCR?
2. In what way IGBT is more advantageous than BJT and MOSFET?
3. Draw the symbol of IGBT.
4. Draw the equivalent circuit of IGBT.
5. What are on state conduction losses? How is it low in IGBT?
6. What is second breakdown phenomenon?
7. What is switching speed of IGBT?
8. Can we observe the transfer and collector characteristics of IGBT on CRO?

POST LAB QUESTIONS:

1. What are merits of IGBT?
2. What are demerits of IGBT?
3. What are the applications of IGBT's?
4. Why silicon used in all power semiconductor devices and why not? Germanium?
5. What is pinch off voltage?
6. What is threshold voltage?

EXPERIMENT – 2

SINGLE PHASE HALF CONTROLLED BRIDGE CONVERTER

Introduction

The experiment on the "Single Phase Half Controlled Bridge Converter" is a critical study in power electronics, particularly in the context of controlled rectification. This converter configuration is widely used in various applications, such as in motor drives, power supplies, and speed control of DC motors. A single phase half controlled bridge converter typically consists of two thyristors (SCRs) and two diodes arranged in a bridge configuration. This setup allows for control over the output voltage by varying the firing angle of the SCRs, providing a means to convert AC to controlled DC output.

The half-controlled bridge has a distinct advantage in terms of component count and cost over fully controlled bridges, but it offers less flexibility in control. In this experiment, students will understand the operational principles of the converter, including the process of phase-controlled rectification, and learn to manipulate the firing angle to control the output voltage.

Learning Objectives

1. Understanding Converter Operation: Gain a comprehensive understanding of the operational principles of the single-phase half controlled bridge converter, including the role of SCRs and diodes in the circuit.

2. Circuit Construction and Design: Learn to construct the half controlled bridge converter circuit, focusing on component selection and understanding the bridge configuration.

3. Firing Angle Control and Its Impact: Develop skills in controlling the firing angle of the SCRs and understand how this affects the output voltage and overall performance of the converter.

4. Analysis of Output Voltage Characteristics: Measure and analyze the output voltage waveform under different firing angles, understanding the relationship between firing angle, output voltage, and load characteristics.

5. Exploring Ripple and Harmonics: Study the ripple content and harmonic distortion in the output voltage, and learn about methods to mitigate these effects.

6. Safety in High-Power Circuit Experiments: Emphasize the importance of safety practices when working with high-power electronic circuits, particularly those involving high voltages.

7. Real-World Applications: Recognize the practical applications of single-phase half controlled bridge converters in industry and consumer electronics.

8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by troubleshooting issues in the converter circuit and making adjustments for optimal performance.

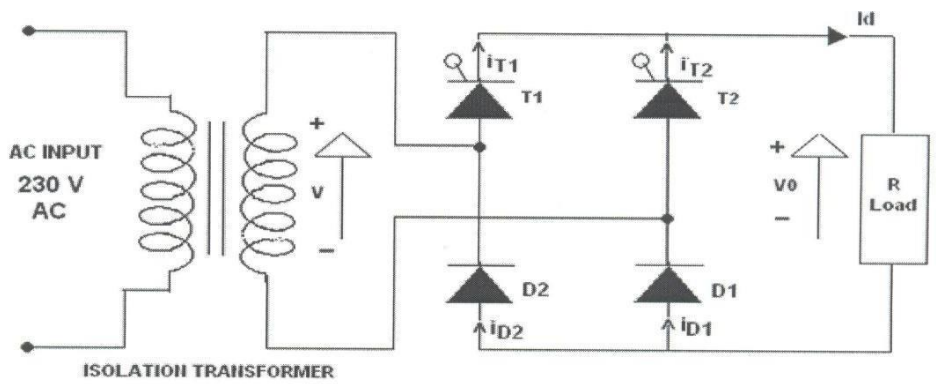
9. Data Analysis and Interpretation: Cultivate the ability to analyze experimental data, compare it with theoretical predictions, and make informed conclusions about the converter's behavior.

By mastering these objectives, students will not only gain a theoretical understanding of single-phase half controlled bridge converters but also acquire practical skills essential for working with and designing controlled rectification systems in various power electronic applications.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Single phase half controlled bridge converter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Isolation Transformer			
5	Variable Rheostat			
6	Inductor			
7	DC Voltmeter			
8	DC Ammeter			

CIRCUIT DIAGRAM:



Circuit Diagram of Single Phase Half Controlled Bridge Converter

PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Connect first 30V AC supply from Isolation Transformer to circuit.
3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.

5. Connect CRO probes and observe waveforms in CRO, Ch-1 or Ch-2, across load and device in single phase half controlled bridge converter.
6. By varying firing angle gradually up to 180^0 and observe related waveforms.
7. Measure output voltage and current by connecting AC voltmeter & Ammeter.
8. Tabulate all readings for various firing angles.
9. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
10. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
11. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

S. No	**Input Voltage (V_{in})**	**Firing angle in Degrees**	**Output voltage (V_0)**		**Output Current (I_0)**	
			Theoretical	**Practical**	**Theoretical**	**Practical**
1						
2						
3						
4						
5						
6						

MODEL CALCULATIONS:

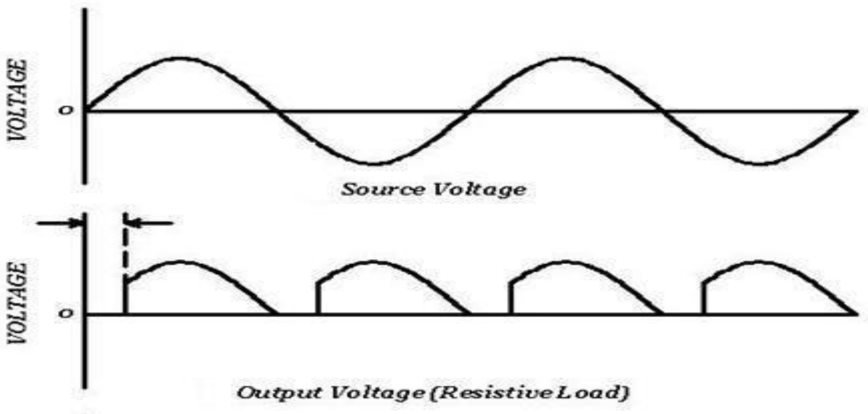
V_0 = $(\sqrt{2}V / \prod) * (1+\text{Cos}\ \alpha)$

I_0 = $(\sqrt{2}V / \prod R) * (1+\text{Cos}$

$\alpha)$ α = Firing Angle

V = RMS Value across transformer output

MODEL GRAPH:



Output Wave Forms of Single Phase Half Controlled Bridge Converter

PRE LAB QUESTIONS:

1. What is the delay angle control of converters?
2. What is natural or line commutation?
3. What is the principle of phase control?
4. What is extinction angle? 5. Can a freewheeling diode be used in this circuit and justify the reason?

POST LAB QUESTIONS:

1. What is conduction angle?
2. What are the effects of adding freewheeling diode in this circuit?
3. What are the effects of removing the freewheeling diode in single phase semi converter?
4. Why is the power factor of semi converters better than that of full converters?
5. What is the inversion mode of converters?

EXPERIMENT – 3

SINGLE PHASE FULLY CONTROLLED BRIDGE CONVERTER WITH R AND RL *LOADS*

Introduction

The experiment on the "Single Phase Fully Controlled Bridge Converter with R and RL Loads" delves into a crucial area of power electronics, focusing on a converter configuration that provides full control over AC to DC conversion. This converter, composed of four thyristors (SCRs) in a bridge arrangement, allows for complete control of the output voltage by adjusting the firing angles of the SCRs. Unlike half-controlled bridges, the fully controlled bridge offers greater flexibility in voltage regulation and is fundamental in applications where precise DC output is necessary, such as in variable speed drives, battery chargers, and power supplies.

This experiment aims to provide a deep understanding of the operation of fully controlled bridge converters and the effect of different load types (resistive and inductive) on their performance. Students will explore how the converter responds under various firing angles and load conditions, and they will learn to analyze the output characteristics in terms of voltage, current, and power factor.

Learning Objectives

1. Understanding Fully Controlled Bridge Operation: Develop a comprehensive understanding of the principles and operation of single-phase fully controlled bridge converters, including the role of SCRs and the importance of firing angle control.
2. Circuit Construction and Load Integration: Learn to construct a fully controlled bridge converter circuit and integrate both resistive (R) and inductive (RL) loads, understanding the impact of load type on converter performance.
3. Firing Angle Control and Output Analysis: Gain expertise in controlling the firing angles of the SCRs and analyze how this affects the output voltage, current waveform, and power factor under different load conditions.
4. Exploring Output Voltage and Current Waveforms: Study the output voltage and current waveforms using oscilloscopes or similar instruments, understanding how load types and firing angles influence these waveforms.
5. Impact of Load on Converter Performance: Investigate the influence of resistive and inductive loads on the performance of the converter, including aspects like ripple, harmonic content, and efficiency.
6. Safety in Power Electronic Experiments: Emphasize the importance of safety practices when working with high-voltage electronic circuits and power electronics equipment.
7. Real-World Application Scenarios: Recognize the practical applications and significance of fully controlled bridge converters in various industrial and electronic systems.
8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by troubleshooting issues in the converter circuit and making adjustments for optimal performance under varying load conditions.

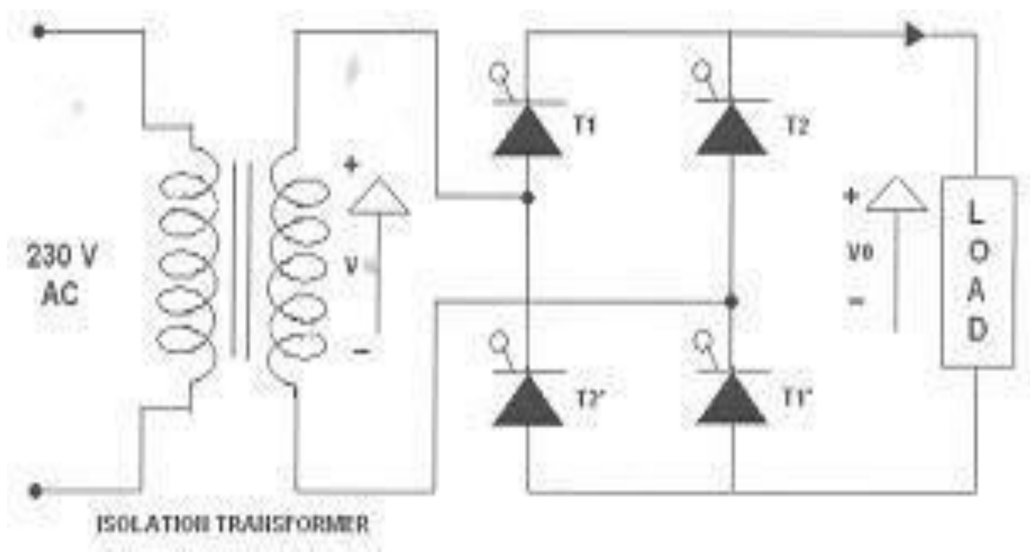
9. Data Analysis and Interpretation: Develop the ability to analyze experimental data, compare it with theoretical predictions, and understand the converter's behavior in different scenarios.

By completing this experiment, students will not only acquire a solid theoretical foundation in fully controlled bridge converters but will also gain practical skills essential for designing and analyzing power electronic systems in various applications.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Single phase full controlled bridge converter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Isolation Transformer			
5	Variable Rheostat			
6	Inductor			
7	DC Voltmeter			
8	DC Ammeter			

CIRCUIT DIAGRAM:



PROCEDURE:

1. Make all connections as per the circuit diagram.

2. Connect firstly 30V AC supply from Isolation Transformer to circuit.
3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.
5. Connect CRO probes and observe waveforms in CRO across load and device in single phase fully controlled bridge converter.
6. By varying firing angle gradually up to 180^0 and observe related waveforms.
7. Measure output voltage and current by connecting AC voltmeter & Ammeter.
8. Tabulate all readings for various firing angles.
9. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
10. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
11. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

S. No	Input Voltage (V_{in})	Firing angle in Degrees	Output voltage (V_0)		Output Current (I_0)	
			Theoretical	Practical	Theoretical	Practical
1						
2						
3						
4						
5						
6						

MODEL CALCULATIONS:

For R-L Load:

$V_0 = (2\sqrt{2}V/\Pi) * \text{Cos}\ \alpha$

$I_0 = (2\sqrt{2}V/\Pi R) * \text{Cos}\ \alpha$

α = Firing Angle

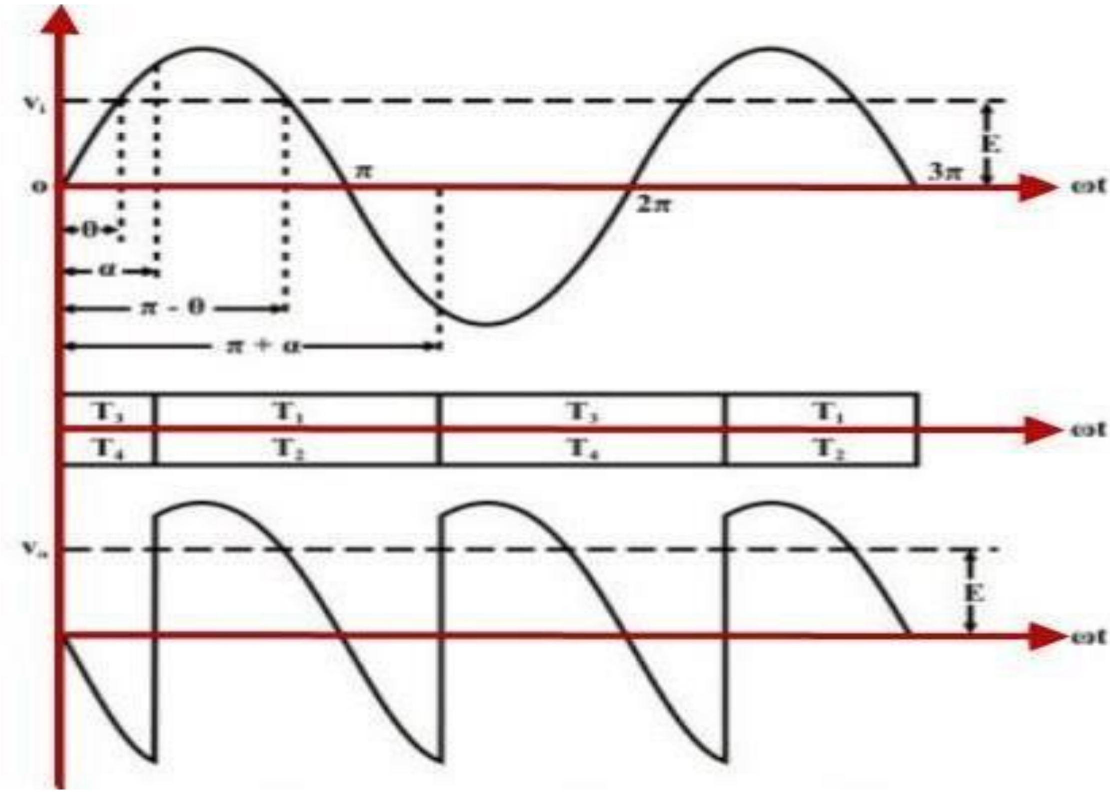
V = RMS Value across transformer output

For R Load:

$V_0 = (\sqrt{2}V/\Pi) * (1+\text{Cos}\ \alpha)$

$I_0 = (\sqrt{2}V/\Pi R) * (1+\text{Cos}\alpha)$

MODEL GRAPH:



Single Phase Fully Controlled Bridge Converter

PRE LAB VIVA QUESTIONS:

1. What will happen if the firing angle is greater than 90 degrees?
2. What are the performance parameters of rectifier?
3. What is the difference between half wave and full wave rectifier?

POST LAB VIVA QUESTIONS:

1. If firing angle is greater than 90 degrees, the inverter circuit formed is called as?
2. What is DC output voltage of single phase full wave controller?

EXPERIMENT – 4

THREE PHASE HALF CONTROLLED BRIDGE CONVERTER WITH R LOAD

Introduction

The experiment on the "Three Phase Half Controlled Bridge Converter with R Load" is an essential exploration in the domain of power electronics, focusing on a significant converter configuration used in industrial and commercial applications. A three-phase half controlled bridge converter comprises a combination of thyristors (SCRs) and diodes arranged in a bridge configuration to convert three-phase AC power to DC power. This particular setup, involving a resistive load (R), is crucial for understanding the basic principles of three-phase rectification and the control of power flow.

This configuration is especially relevant in scenarios where intermediate control of the output voltage is required, such as in certain types of motor drives, heating applications, and power regulation systems. The experiment is designed to provide insight into the operational characteristics of three-phase half controlled bridge converters and to understand the impact of controlled rectification in three-phase systems.

Learning Objectives

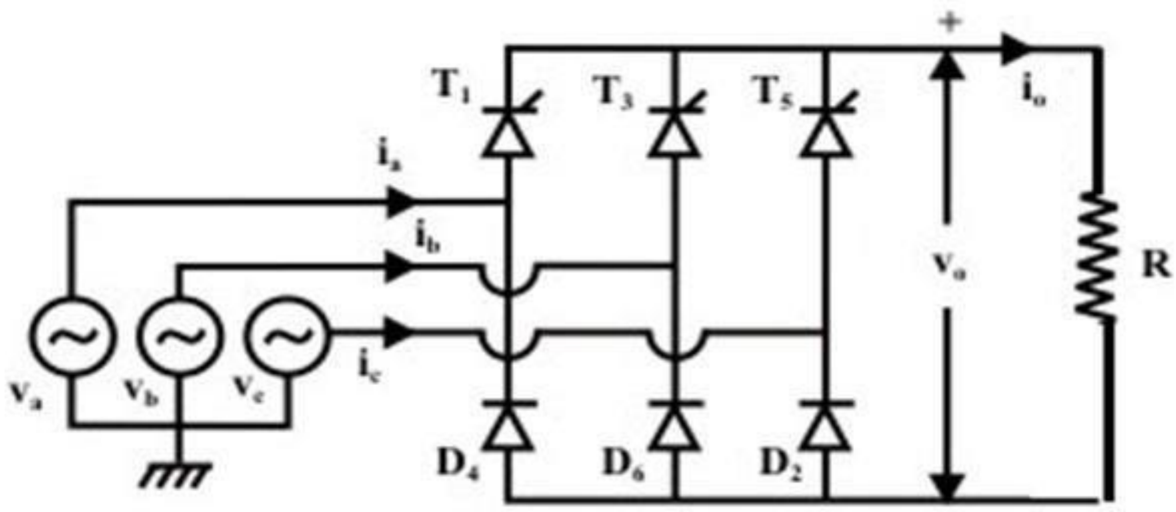
1. Understanding Three-Phase Half Controlled Bridge Operation: Develop a deep understanding of the operational principles of three-phase half controlled bridge converters, including the configuration of SCRs and diodes.
2. Circuit Construction and Design: Learn to construct a three-phase half controlled bridge converter circuit and integrate a resistive load, focusing on the proper arrangement and connection of components.
3. Analysis of Output Voltage and Current: Gain expertise in analyzing the output voltage and current waveforms of the converter, understanding the impact of the firing angle of SCRs on these parameters.
4. Exploring the Effect of Firing Angle Control: Study how varying the firing angle of SCRs influences the output voltage, particularly in terms of rectification and ripple characteristics.
5. Safety in High-Power Circuit Experiments: Emphasize the importance of safety practices when working with high-voltage and high-power electronic circuits.
6. Measurement Techniques and Instrumentation: Develop skills in using appropriate measurement instruments, such as oscilloscopes and multimeters, to observe and analyze the converter's performance.
7. Real-World Application Scenarios: Recognize the practical applications of three-phase half controlled bridge converters in industrial settings and power systems.
8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by troubleshooting issues in the converter circuit and making adjustments for optimal performance.
9. Data Analysis and Interpretation: Cultivate the ability to interpret experimental data, compare it with theoretical predictions, and draw conclusions about the converter's behavior under various operating conditions.

By mastering these objectives, students will not only learn the theoretical aspects of three-phase half controlled bridge converters but will also acquire practical skills essential for designing and analyzing power electronic systems in various applications.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Three phase half controlled bridge converter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Three phase transformer			
5	Rheostat			
6	DC Voltmeter			
7	DC Ammeter			

CIRCUIT DIAGRAM:



Half Controlled bridge converter with R load

PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Connect firstly 3 phase AC supply from three phase transformer to circuit.
3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.

5. Connect CRO probes and observe waveforms in CRO across load and device in three phase half controlled bridge converter.
6. By varying firing angle gradually up to 180^0 and observe related waveforms.
7. Measure output voltage and current by connecting DC voltmeter & Ammeter.
8. Tabulate all readings for various firing angles.
9. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

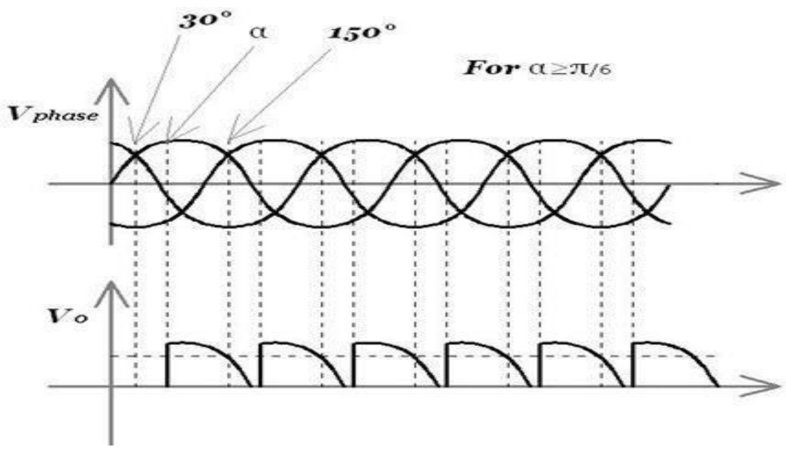
S. No	Input Voltage (V_{in})	Firing Angle in Degrees	Output voltage (V_0)		Output Current (I_0)	
			Theoretical	Practical	Theoretical	Practical
1						
2						
3						
4						
5						
6						

MODEL CALCULATIONS:

Vo = 3 Vml*(1+cosα)/2π Io = 3 Vml*(1+cosα)/2πR α= firing angle

Vml = line to line voltage

MODEL GRAPHS:



Input and output wave forms of a three phase half controlled bridge converter

PRE LAB QUESTIONS

1. A converter which can operate in both 3 pulse and six pulse modes is?
2. What is the interval for SCRs triggering in three phase semi converter?
3. What is the interval for SCRs triggering in three phase full converter?
4. What is the function of freewheeling diode in three phase converters?

POST LAB QUESTIONS

5. What are the advantages of three phase half controlled converters?
6. Which quadrant operation is possible with three phase half controlled converter?

EXPERIMENT – 5

SINGLE PHASE A.C. VOLTAGE CONTROLLER

Introduction

The experiment on the "Single Phase AC Voltage Controller" is a fundamental study in power electronics, focusing on the control and modulation of AC voltage using power electronic devices. AC voltage controllers, also known as AC regulators, are used to adjust the RMS (Root Mean Square) value of the alternating voltage applied to a load directly from an AC supply. This is typically achieved through methods like phase control or on-off control using devices like thyristors, TRIACs, or SCRs. These controllers are widely used in applications such as light dimming, speed control of AC motors, and temperature control in heating applications.

The experiment involves setting up a single-phase AC voltage controller circuit and examining how the application of different firing angles or duty cycles affects the output voltage supplied to a load. This provides practical insights into the principles of AC voltage regulation and the behavior of controllers under various load conditions.

Learning Objectives

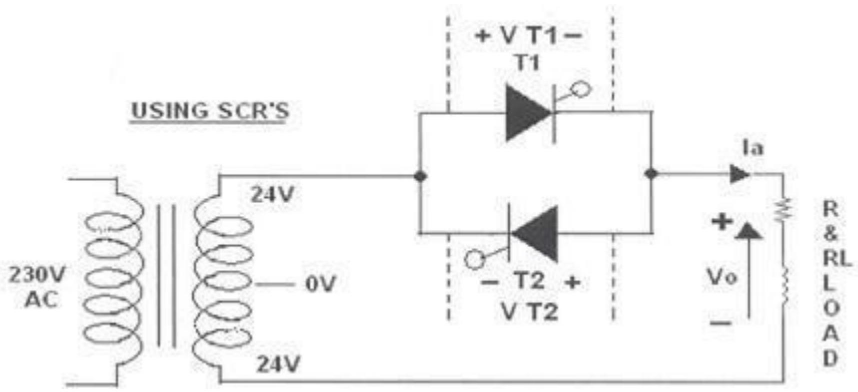
1. Understanding AC Voltage Controller Principles: Gain a comprehensive understanding of the operational principles of single-phase AC voltage controllers, including their basic configuration and the role of power electronic devices in controlling AC voltage.
2. Circuit Construction and Design: Learn to construct a single-phase AC voltage controller circuit, focusing on the selection and implementation of control devices like thyristors or TRIACs.
3. Analysis of Voltage Control Techniques: Develop skills in applying different voltage control techniques, such as phase control or integral cycle control, and analyzing their impact on the output voltage waveform.
4. Exploring the Impact of Firing Angle and Duty Cycle: Study how the firing angle in phase control or the duty cycle in on-off control influences the RMS output voltage and power delivered to the load.
5. Measurement and Interpretation of Waveforms: Use instruments like oscilloscopes to measure and interpret the output voltage waveforms under different control settings, understanding the waveform characteristics and their implications on load performance.
6. Safety in Electronic Circuit Experiments: Emphasize the importance of safety when handling and testing high-voltage AC circuits.
7. Real-World Application Scenarios: Recognize the practical applications of single-phase AC voltage controllers in various domains, such as motor speed control, lighting control, and heating systems.
8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by diagnosing and troubleshooting issues in the voltage controller circuit and optimizing its performance for specific applications.
9. Data Analysis and Interpretation: Develop the ability to analyze experimental data, compare it with theoretical expectations, and understand the performance and limitations of AC voltage controllers.

By completing this experiment, students will not only learn the theoretical concepts of AC voltage control but will also acquire practical skills essential for designing and analyzing AC voltage control systems for various electronic applications.

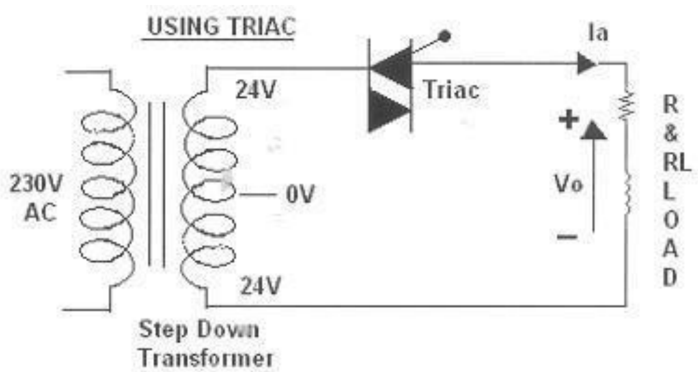
APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Single phase AC voltage controller power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Isolation Transformer			
5	Variable Rheostat			
6	Inductor			
7	AC Voltmeter			
8	AC Ammeter			

CIRCUIT DIAGRAM:



Single Phase AC Voltage Controller with Thyristors



Single Phase AC Voltage Controller with Traic

PROCEDURE:

AC VOLTAGE CONTROLLER WITH TWO THYRISTORS:

1. Make all connections as per the circuit diagram.
2. Connect firstly 30V AC supply from Isolation Transformer to circuit.

3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.
5. Observe waveforms in CRO, across load by varying firing angle gradually up to 180^0.
6. Measure output voltage and current by connecting AC voltmeter & Ammeter.
7. Tabulate all readings for various firing angles.
8. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
9. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
10. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

A.C. VOLTAGE CONTROLLER WITH TRIAC:

1. Make all connections as per the circuit diagram.
2. Connect firstly 30V AC supply from Isolation Transformer to circuit.
3. Connect firing pulse from firing circuit to TRIAC as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.
5. Observe waveforms in CRO, across load by varying firing angle gradually up to 180^0.
6. Measure output voltage and current by connecting AC voltmeter & Ammeter.
7. Tabulate all readings for various firing angles.
8. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
9. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
10. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

S. No.	Input Voltage (V_{in})	Firing angle in Degrees	Output voltage (V_{0r})		Output Current (I_{0r})	
			Theoretical	Practical	Theoretical	Practical
1						
2						
3						
4						
5						
6						

MODEL CALCULATIONS:

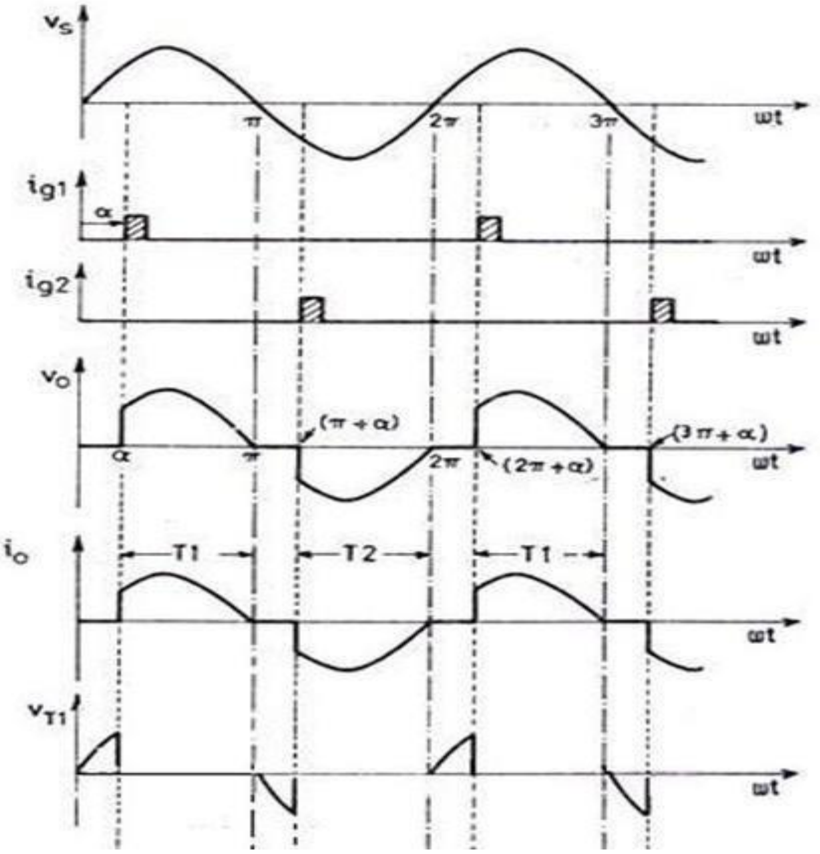
I0r = V0r / R α

= Firing

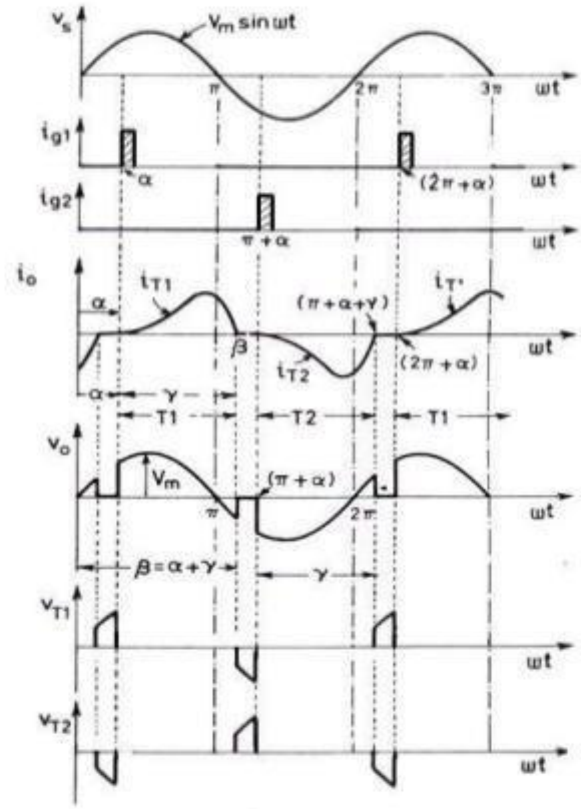
Angle

V = RMS Value across transformer output

MODEL GRAPH:



Single Phase AC Voltage controller with R - Load



Single Phase AC voltage controller with RL Load

PRE LAB QUESTIONS:

1. Why should the two trigger sources be isolated?
2. What are the advantages and the disadvantages of phase control?
3. What is phase control?

POST LAB QUESTIONS:

1. What type of commutation is used in this circuit?
2. What are the effects of load inductance on the performance of AC voltage controllers?
3. What is extinction angle?

EXPERIMENT – 6
SINGLE PHASE CYCLO - CONVERTER WITH R AND RL LOADS

Introduction

The experiment on the "Single Phase Cycloconverter with R and RL Loads" focuses on a sophisticated power electronics device used to convert a fixed-frequency AC input into a variable-frequency AC output without an intermediate DC link. Cycloconverters are particularly significant in applications that require low-frequency output, such as in driving large AC motors for industrial processes like rolling mills, mine hoists, and cement kilns.

This type of converter directly converts the input AC waveform to a lower frequency AC waveform by controlling the firing angles of power electronic switches like thyristors. The experiment with resistive (R) and resistive-inductive (RL) loads provides insights into how the cycloconverter handles different load types and the implications for output waveform quality and power factor.

Learning Objectives

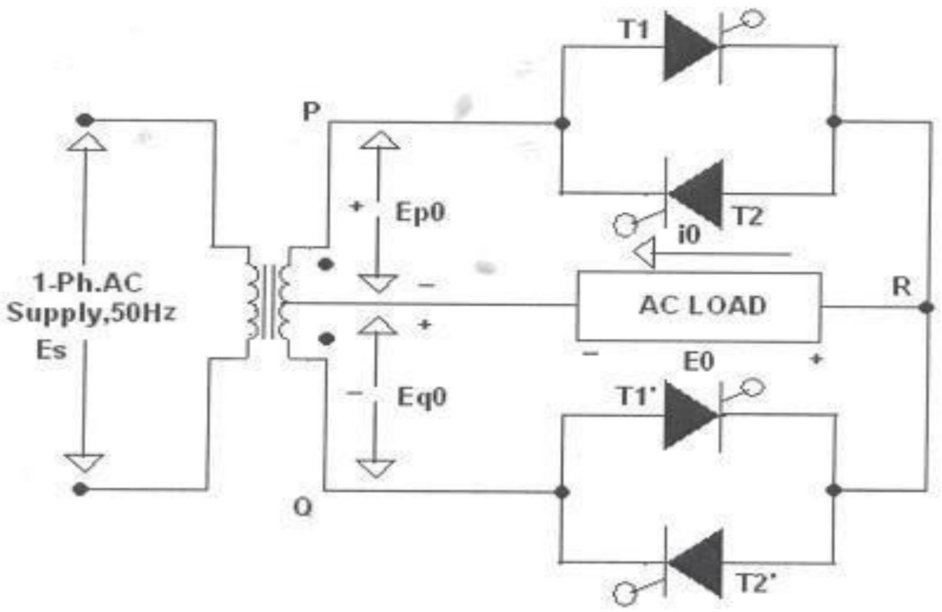
1. Understanding Cycloconverter Principles: Develop a comprehensive understanding of cycloconverter operation, including its ability to generate variable-frequency output from a fixed-frequency input.
2. Circuit Construction and Control Strategy: Learn to construct a single-phase cycloconverter circuit and understand the control strategies for firing the power electronic switches to achieve the desired frequency conversion.
3. Output Frequency and Waveform Analysis: Gain skills in analyzing the output frequency and waveform of the cycloconverter, understanding how it varies with the control strategy and load types.
4. Impact of Load Types on Performance: Study how resistive and resistive-inductive loads impact the performance of the cycloconverter, including effects on output waveform, harmonic content, and power factor.
5. Measurement Techniques: Develop proficiency in using measurement tools like oscilloscopes and frequency meters to measure the output frequency and waveform characteristics of the cycloconverter.
6. Exploring Harmonics and Distortion: Understand the generation of harmonics and distortion in the output waveform and learn about methods to mitigate these effects.
7. Safety in High-Power Electronic Experiments: Emphasize the importance of safety practices when dealing with high-power electronic circuits.
8. Real-World Application Scenarios: Recognize the practical applications and significance of cycloconverters in industries that require variable-frequency AC drives.
9. Problem-Solving and Critical Thinking: Enhance problem-solving skills by troubleshooting issues in the cycloconverter circuit and optimizing its performance for various loads.
10. Data Analysis and Interpretation: Develop the ability to analyze experimental data, compare it with theoretical expectations, and draw conclusions about the performance of the cycloconverter.

By mastering these objectives, students will acquire not only a theoretical understanding of cycloconverters but also practical skills essential for designing and analyzing variable-frequency AC power systems in industrial applications.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Single phase Cycloconverter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Isolation Transformer (Centre - Tapped)			
5	Variable Rheostat			
6	Inductor			
7	AC Voltmeter			
8	AC Ammeter			

CIRCUIT DIAGRAM :



Circuit Diagram of Single Phase Cyclo - Converter

PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Connect firstly (30V-0-30V) AC supply from Isolation Transformer to circuit.
3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.

4. Connect resistive load 200Ω / 5A to load terminals.
5. Set the frequency division switch to (2,3,4,…9) your required output frequency.
6. Switch ON the MCB and IRS switch and trigger output ON switch.
7. Observe waveforms in CRO, across load by varying firing angle gradually up to 180^0 and also for various frequency divisions (2,3,4,…9).
8. Measure output voltage and current by connecting AC voltmeter & Ammeter.
9. Tabulate all readings for various firing angles.
10. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
11. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
12. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

S. No	Input Voltage (V_{in})	Firing angle in Degrees	Frequency Division	V_o (V)	I_o (A)	Input frequency f_s	Output frequency f_o	f_o/f_s

MODEL CALCULATIONS:

V_{0r} =

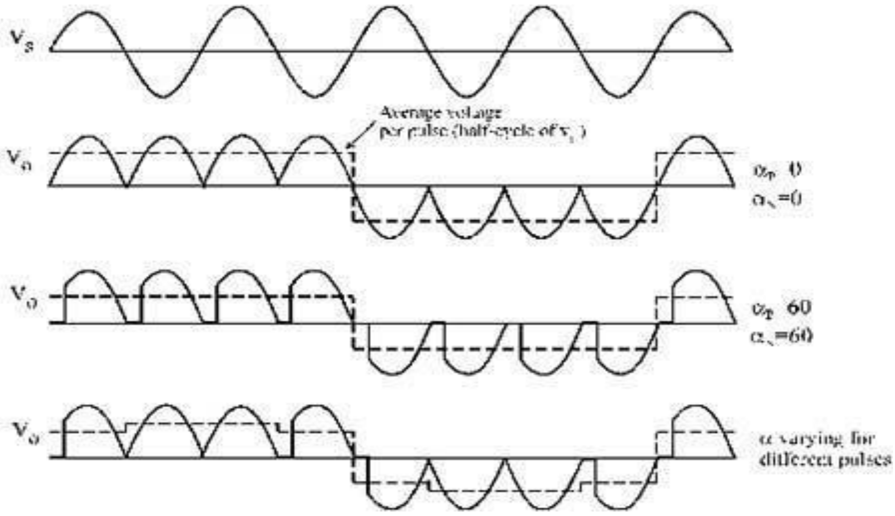
I_{0r} = V_{0r} / R θ

= Firing

Angle

V = RMS Value across transformer output

MODEL GRAPH:



Output Wave Forms of Single Phase Cyclo – Converter

PRE LAB QUESTIONS:

1. What is meant by Cyclo-converter? What are the types of Cyclo-converters?
2. Classify Cyclo converters.
3. What is step up Cyclo-converter & step down cyclo - converter ?
4. Differentiate step-down cycloconverter and step-up cycloconverter.
5. Why forced commutation circuit is employed in case of cyclo inverter?

POST LAB QUESTIONS:

1. What are the Applications of Cycloconverter?
2. What is meant by Positive & negative converter groups in cycloconverter?
3. List the applications of cycloconverter.
4. List the advantages & disadvantages of cycloconverters.

EXPERIMENT 7

SINGLE PHASE SERIES INVERTER WITH R AND RL LOADS

Introduction

The experiment on the "Single Phase Series Inverter with R and RL Loads" delves into an essential aspect of power electronics, focusing on the conversion of DC power to AC power using a series inverter configuration. A series inverter, also known as a series resonant inverter, operates by creating an oscillating current through a series LC circuit, which, when applied to a load, results in an AC output. This type of inverter is particularly efficient at generating AC power at high frequencies and is used in applications like induction heating, wireless power transfer, and high-frequency AC power supply systems.

In this experiment, students will construct a single-phase series inverter and explore its operation with different types of loads - resistive (R) and resistive-inductive (RL). They will investigate how the inverter responds to changes in load conditions and understand the importance of tuning the resonant frequency of the LC circuit to match the desired output frequency.

Learning Objectives

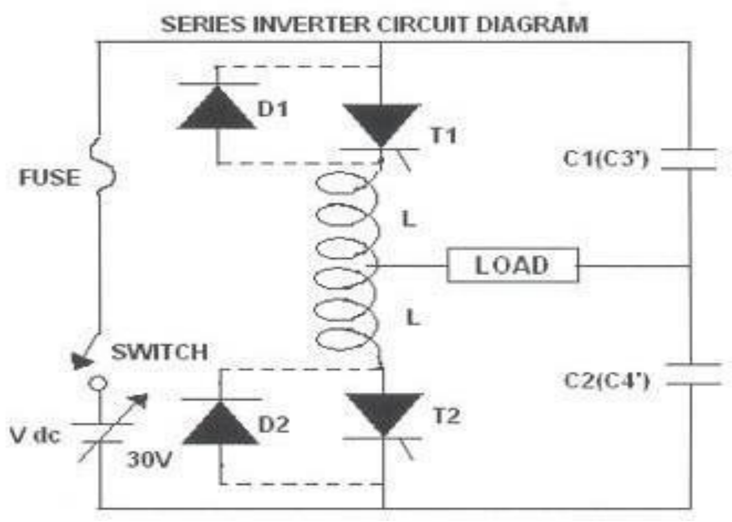
1. Understanding Series Inverter Principles: Develop a solid understanding of the operational principles of single-phase series inverters, including the role of the LC resonant circuit in generating AC output.
2. Circuit Construction and Load Integration: Learn to construct a single-phase series inverter circuit and integrate both resistive (R) and resistive-inductive (RL) loads, understanding the impact of these loads on inverter performance.
3. Analysis of Output Voltage and Frequency: Develop skills in analyzing the output voltage waveform and frequency of the inverter, understanding how these parameters are influenced by the resonant frequency of the LC circuit and the load characteristics.
4. Resonant Frequency Tuning and Optimization: Gain experience in tuning the resonant frequency of the LC circuit to achieve optimal performance, particularly in terms of output voltage stability and waveform quality.
5. Exploring the Effect of Load Variations: Study how varying the load from purely resistive to resistive-inductive affects the inverter's efficiency, output voltage waveform, and power factor.
6. Safety in High-Frequency Electronic Experiments: Emphasize the importance of safety when working with high-frequency circuits and high-voltage inverters.
7. Measurement Techniques and Instrumentation: Develop proficiency in using instruments like oscilloscopes to observe and analyze the inverter's output waveform and frequency.
8. Real-World Application Scenarios: Understand the practical applications of single-phase series inverters in various electronic and industrial systems.
9. Problem-Solving and Critical Thinking: Enhance problem-solving skills by diagnosing and troubleshooting issues in the series inverter circuit and making adjustments for optimal performance.
10. Data Analysis and Interpretation: Cultivate the ability to analyze experimental data, compare it with theoretical predictions, and understand the performance characteristics of the series inverter under different load conditions.

By completing this experiment, students will not only gain a theoretical understanding of single-phase series inverters but will also acquire practical skills crucial for designing and analyzing high-frequency power conversion systems.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Series inverter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Regulated dc power supply			
5	Variable Rheostat			
6	Inductor			

CIRCUIT DIAGRAM:



Circuit Diagram Single Phase Series Inverter

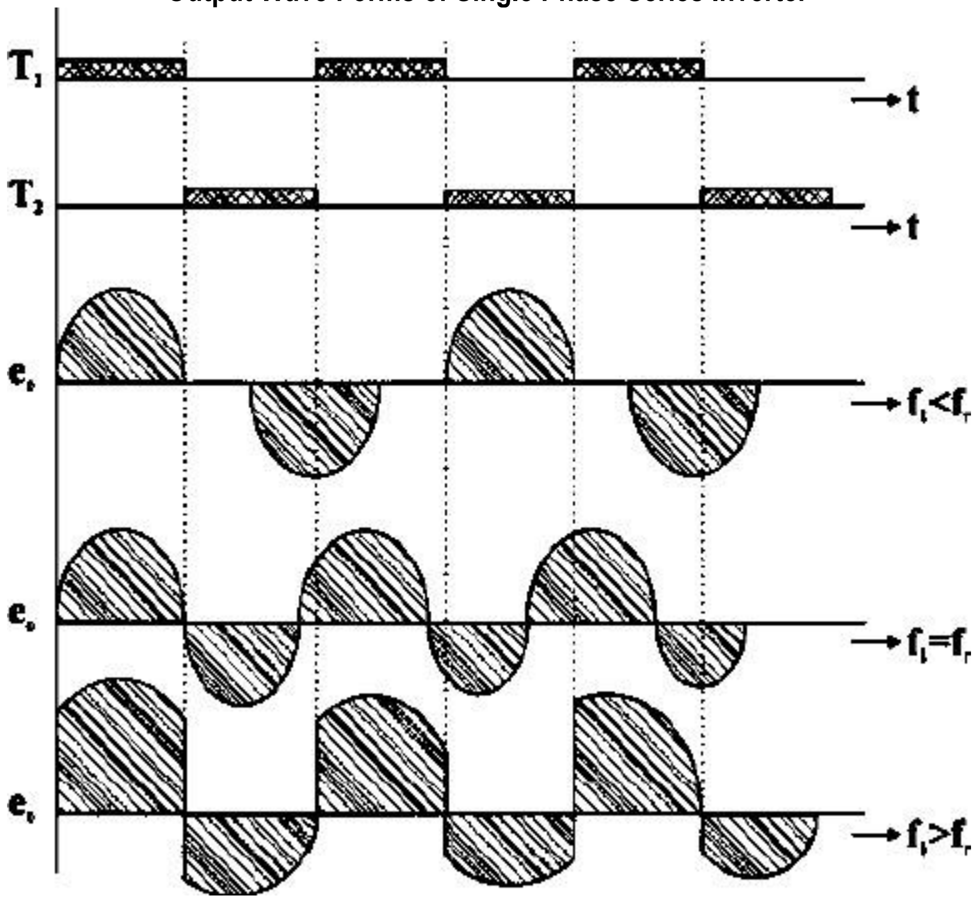
PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Give the DC power supply 30V to the terminal pins located in the power circuit.
3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.
5. By varying the frequency pot, observe related waveforms.

6. If the inverter frequency is increases above the resonant frequency of the power circuit commutation fails. Then switch OFF the DC supply, reduce the inverter frequency and try again.
7. Repeat the above same procedure for different value of L,C load and also above the wave forms with and without fly wheel diodes.
8. Total output wave forms entirely depends on the load, and after getting the perfect wave forms increase the input supply voltage up to 30V and follow the above procedure.
9. Switch OFF the DC supply first and then Switch OFF the inverter.(Switch OFF the trigger pulses will lead to short circuit)

MODEL WAVEFORMS:

Output Wave Forms of Single Phase Series Inverter



PRE LAB QUESTIONS:

1. Why is this circuit called as series inverter?
2. What is the type of commutation for series inverter?
3. What is the configuration of inductor?
4. What is the principle of series inverter?
5. Disadvantages of series inverter.
6. On what principle series inverter works?

POST LAB QUESTIONS:

1. What is the dead zone of an inverter?
2. Up to what maximum voltage will the capacitor charge during circuit operation?
3. What is the amount of power delivered by capacitor? 4. What is the purpose of coupled inductors in half bridge resonant inverters?

EXPERIMENT – 8

DC JONE'S CHOPPER

Introduction

The experiment on the "DC Jones Chopper" is a pivotal study in the field of power electronics, focusing on a specific type of DC-to-DC converter. A Jones Chopper is a classical chopper circuit known for its simplicity and efficiency in controlling the DC output voltage from a fixed DC input voltage. This control is achieved by rapidly switching the DC input on and off, thereby changing the average voltage across the load. Jones Choppers are commonly used in various applications such as speed control in DC motors, regenerative braking systems, and battery-operated devices.

In this experiment, students will construct a DC Jones Chopper circuit and explore its operational characteristics. They will learn about the principles of rapid switching, duty cycle control, and the impact of these factors on the output voltage and current. The experiment offers practical insights into the design and function of choppers and their role in modern power electronic systems.

Learning Objectives

1. Understanding DC Chopper Principles: Gain a comprehensive understanding of the operational principles of DC choppers, specifically the Jones Chopper, including the concept of duty cycle control for voltage regulation.

2. Circuit Construction and Design: Learn to construct a DC Jones Chopper circuit, focusing on component selection and understanding the role of each component in the chopping action.

3. Analysis of Duty Cycle and Output Voltage: Develop skills in varying the duty cycle of the chopper and analyzing its impact on the output voltage and current, understanding the relationship between duty cycle, load, and output characteristics.

4. Exploring Switching Devices and Techniques: Study the use of different switching devices such as thyristors, MOSFETs, or IGBTs in the chopper circuit and the techniques for controlling them.

5. Measurement and Interpretation of Waveforms: Use instruments like oscilloscopes to measure and interpret the voltage and current waveforms in the chopper circuit, focusing on the effects of chopping frequency and duty cycle.

6. Safety in High-Power Electronic Experiments: Emphasize the importance of safety practices when working with high-power and high-frequency electronic circuits.

7. Real-World Application Scenarios: Recognize the practical applications of DC choppers in various domains, such as in electric vehicles, renewable energy systems, and industrial motor control.

8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by diagnosing and troubleshooting issues in the chopper circuit and optimizing its performance for specific applications.

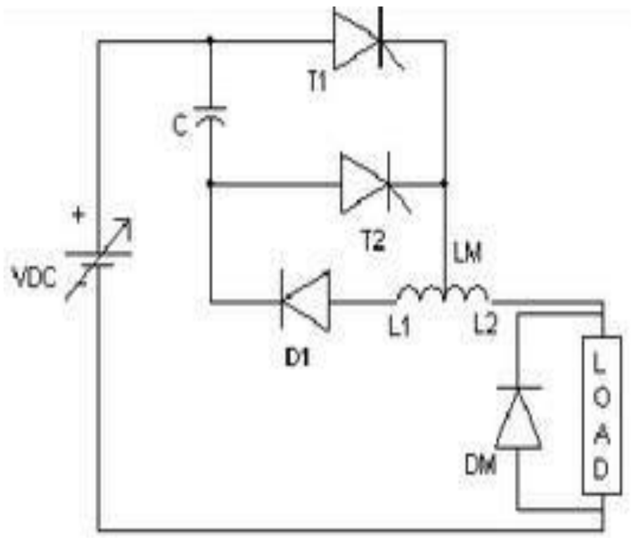
9. Data Analysis and Interpretation: Cultivate the ability to analyze experimental data, compare it with theoretical expectations, and draw conclusions about the performance of the Jones Chopper.

By mastering these objectives, students will not only learn the theoretical aspects of DC Jones Choppers but will also acquire practical skills essential for designing and analyzing DC-to-DC conversion systems in various electronic and electrical applications.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	DC chopper power MODEL			
2	Triggering circuit (DC chopper)			
3	Rheostat			
4	Digital multimeter			
5	CRO			
6	Patch Cards			

CIRCUIT DIAGRAM:



Circuit Diagram of Jones Chopper

T1, T2 – TYN
616 D1 – BYQ
28200
C – Commutation Capacitor 10μF / 100V
L1- 0 –L2 - Commutation Inductor 500-0-500μH / 2A

PROCEDURE:

a) For R – Load:

1. Connections are made as shown in the figure. Use 50Ώ Rheostat for R - Load (Freewheeling diode (DM) is to be connected only for RL load).
2. Adjust V_{RPS} output to 10v and connect to DC chopper MODEL.
3. Switch on DC toggle switch of chopper MODEL.
4. Switch on the trigger input by pushing- in pulse switch.
5. Observe the output waveform across load on CRO.
6. Keep the duty cycle at mid position and vary the frequency from minimum to maximum and record the output voltage readings.
7. Note down the output waveform for mid value of frequency and duty cycle. **b) R - L Load:**

1. Connections are made as shown in fig. Load is 50Ώ Rheostat in series with inductor L =25mH or 50mH.
2. Follow the same procedure as listed in steps 2 to 8 above.3. Readings and output waveform is to be recorded with and without freewheeling diode. [NOTE: In both switching on / switching off of the equipment. First use DC toggle switch and then the pulsar].

TABULAR COLUMN:

Constant Duty Cycle

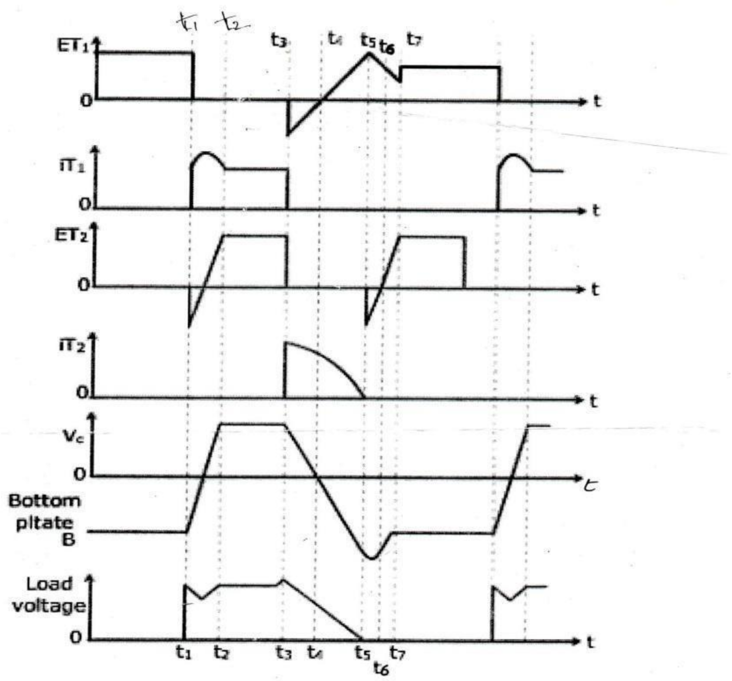
Duty Cycle: 50%, V_{IN}=10 to 15 V

S. No	Frequency(Hz)	V0(Volts)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Constant Frequency, Frequency Control

S. No	T_{ON}(sec)	T_{OFF}(sec)	Duty Cycle (%)	V_O (Volts)
1				
2				
3				
4				
5				
6				
7				
8				
9				

MODEL GRAPH:



Output Characteristics of DC Jones Chopper

PRE LAB QUESTIONS:

1. What are choppers?
2. What does a chopper consist of? 3. On what basis choppers are classified in quadrant configurations?

POST LAB QUESTIONS:

1. Define duty cycle.
2. How can ripple current be controlled?
3. What is step up chopper?
4. On what does the commutating capacitor value depend on?
5. What are the disadvantages of choppers?
6. How do they have high efficiency?
7. What are the applications of dc choppers?

EXPERIMENT – 9(A)

SIMULATION OF SINGLE PHASE FULL WAVE RECTIFIER USING RLE LOADS

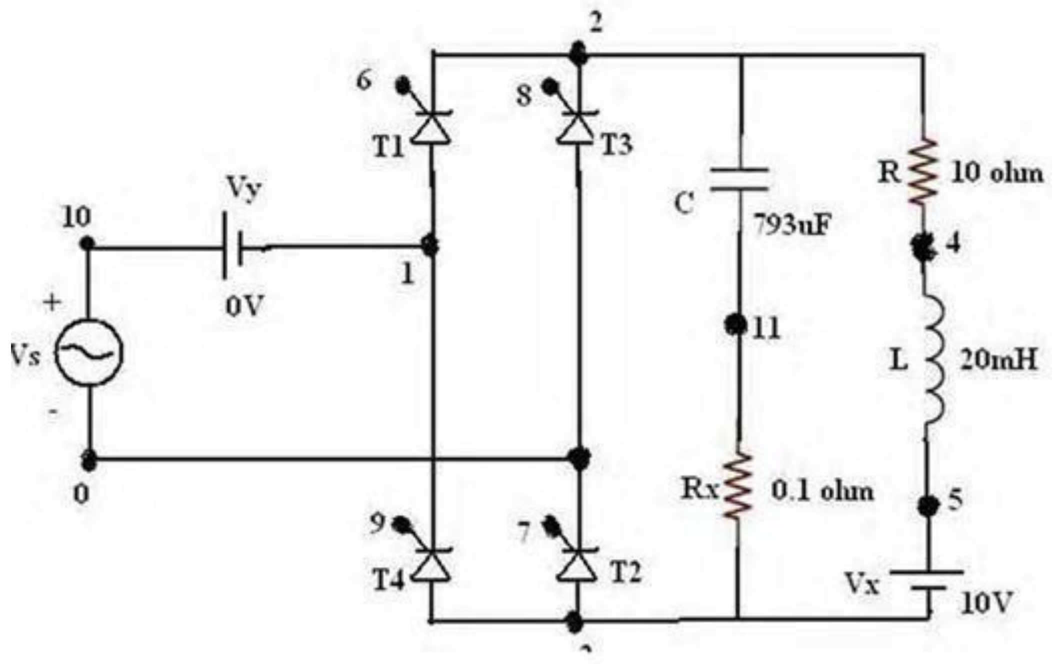
AIM:

To obtain the performance characteristics of Single Phase Semi converter for R, RL, RLE Loads Using MATLAB / Simulink

APPARATUS:

S. No.	Name of the Equipment
1.	PC With Desktop
2.	Matlab / Simulink

CIRCUIT DIAGRAM:



Circuit Diagram of Simulation of Single Phase Full Wave Rectifier

PROCEDURE:

1. Represent the nodes for a given circuit.
2. From desktop of your computer click on “START” menu followed by “programs” and then clicking appropriate multisim

3. Connect in the Simulator the given circuit diagram
4. To make changes in the program open the circuit file, modify, save & Run the program.
5. If there are no errors, simulation will be completed & it will be displayed with a statement as "simulation completed".

PRE LAB QUESTIONS:

1. If we place freewheeling diode what happens in full controlled converters?
2. What is the output voltage equation of a 3-phase fully controlled bridge converter?
3. What are the parameters effects the frequency of the ripple in the output voltage of 3phase semi converter?

POST LAB QUESTIONS:

1. Explain operation of rectifier with RL Load.
2. How to plot rectifier output waveforms?

EXPERIMENT9 (B)

SIMULATION OF SINGLE PHASE AC VOLTAGE CONTROLLER USING RLE LOADS

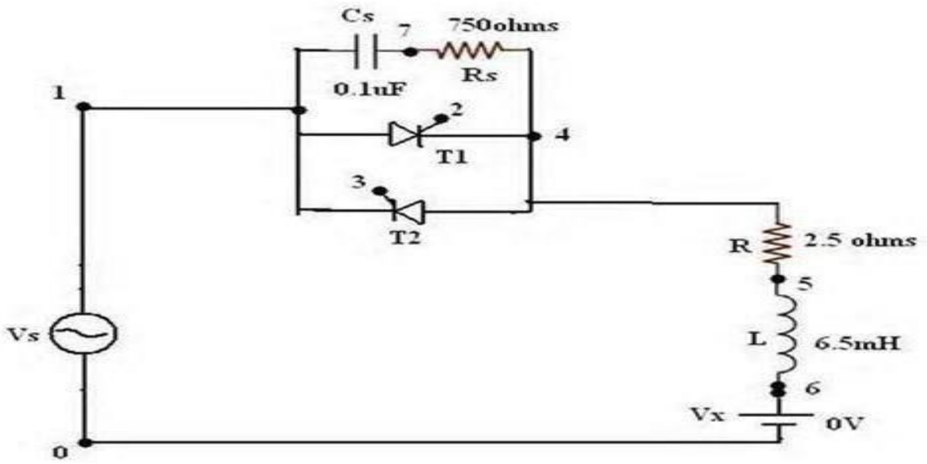
AIM:

To obtain the performance characteristics of Single Phase for R, RL, RLE Loads Using MATLAB / Simulink

APPARATUS:

S. No.	Name of the Equipment
1.	PC With Desktop
2.	MATLAB / Simulink

CIRCUIT DIAGRAM:



Circuit Diagram of PSPICE Simulation of Single Phase Ac Voltage Controller

PROCEDURE:

1. Represent the nodes for a given circuit.
2. From desktop of your computer click on “START” menu followed by “programs” and then clicking appropriate multisim
3. Connect in the Simulator the given circuit diagram
4. To make changes in the program open the circuit file, modify, save & Run the program.
5. If there are no errors, simulation will be completed & it will be displayed with a statement as “simulation completed”.

PRE LAB QUESTIONS:

1. What is the output voltage equation of AC Voltage controller?

2. Draw the output waveform of AC voltage controllers with a firing angle of 90 degrees.

POST LAB QUESTIONS:

1. What are the applications of fully controlled rectifiers?
2. What the advantages are of fully controlled over half controlled rectifiers?

EXPERIMENT – 10 (A)
SIMULATION OF RESONANT PULSE COMMUTATION CIRCUIT

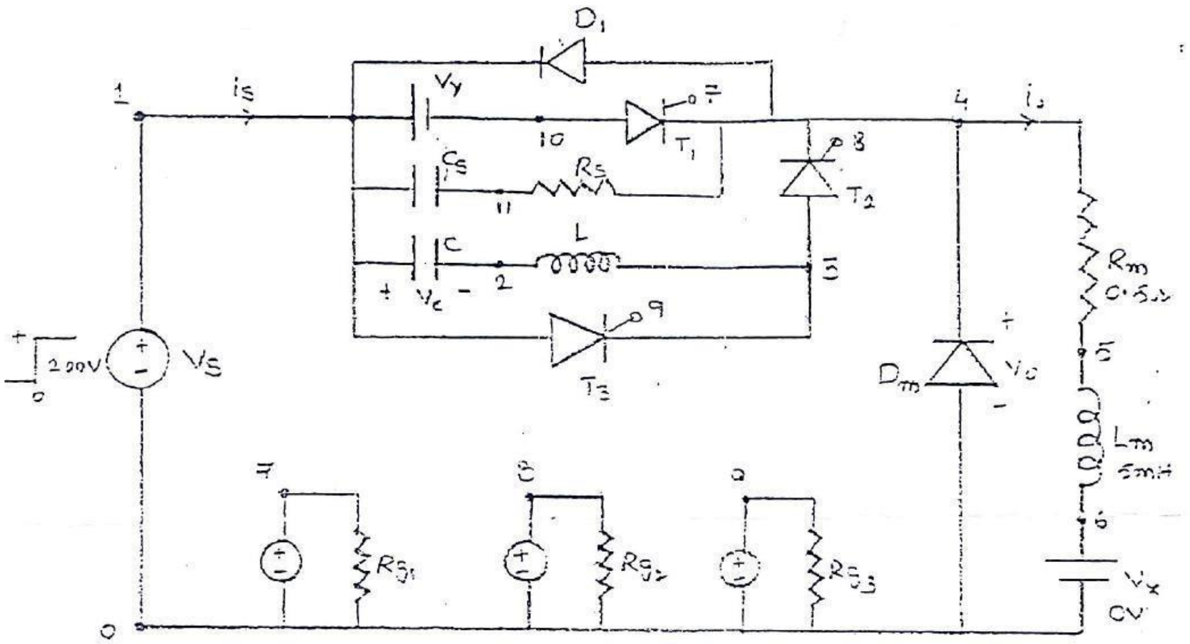
AIM:

To obtain the performance characteristics of a Resonant Pulse Commutation Circuit

APPARATUS:

S. No.	Name of the Equipment
1.	PC With Desktop
2.	MATLAB / Simulink

CIRCUIT DIAGRAM:



Circuit Diagram of PSPICE Simulation of Resonant Pulse Commutation Circuit

PROCEDURE:

6. Represent the nodes for a given circuit.
7. From desktop of your computer click on “START” menu followed by “programs” and then clicking appropriate multisim
8. Connect in the Simulator the given circuit diagram
9. To make changes in the program open the circuit file, modify, save & Run the program.

10. If there are no errors, simulation will be completed & it will be displayed with a statement as “simulation completed”.

PRE LAB QUESTIONS:

1. What is mean by Commutation?
2. What is mean by Resonant Commutation? 3. Explain Resonant Pulse Commutation.

POST LAB QUESTIONS:

1. What are the applications of Resonant pulse Commutation?

EXPERIMENT – 10 (B)

SIMULATION OF BUCK CHOPPER

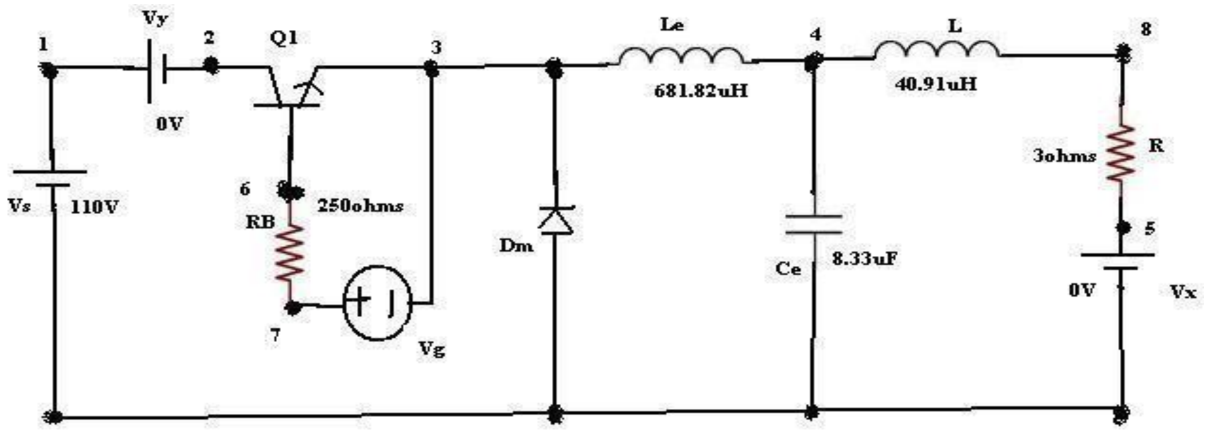
AIM:

To obtain the performance characteristics of BUCK CHOPPER

APPARTUS:

S. No.	Name of the Equipment
1.	PC With Desktop
2.	PSPICE

CIRCUIT DIAGRAM:



Circuit Diagram of PSPICE Simulation of Buck Chopper

PROCEDURE:

1. Represent the nodes for a given circuit.
2. Write spice program by initializing all the circuit parameter as per given flow chart.
3. From desktop of your computer click on " START " menu followed by " programs " and then clicking appropriate program group as " DESIGN LAB EVAL8 followed by "
   DESIGN MANAGER."
4. Open the run text editor from microsim window & start writing pspice program.
5. Save the program with .cir extension.
6. Open the run spice A / D window from microsim window.
7. Open file menu from run spice A / D window then open saved circuit file.

8. If there are any errors, simulates will be displayed with statement as “simulation error occurred”.
9. To see the errors click on o / p file icon and open examine o / p.
10. To make changes in the program open the circuit file, modify, save & Run the program.
11. If there are no errors, simulation will be completed & it will be displayed with a statement as “simulation completed”.
12. To see the o / p click on o / p file icon & open examine o / p then note down the values.
13. If .probe command is used in the program, click on o / p file icon &open run probe. Select variables to plot on graphical window and observe the o / p plots then take print outs of that.

PRE LAB QUESTIONS:

1. What is mean by Buck Chopper?
2. What is the difference between the Buck and Boost Chopper?
3. What are the applications of Buck Chopper?
4. What are the applications of Boost Chopper?

POST LAB QUESTIONS:

1. What is the output voltage equation for step up chopper?
2. Define duty cycle.
3. What is the output voltage equation for step down chopper?
4. What is the output voltage equation for step up / down chopper?

EXPERIMENT – 11

SIMULATION OF SINGLE PHASE INVERTER WITH PWM CONTROL

AIM:

To obtain the performance characteristics of single phase inverter with PWM control.

APPARATUS:

S. No	Name of the Equipment
1.	PC With Desktop
2.	PSPICE

CIRCUIT DIAGRAM:

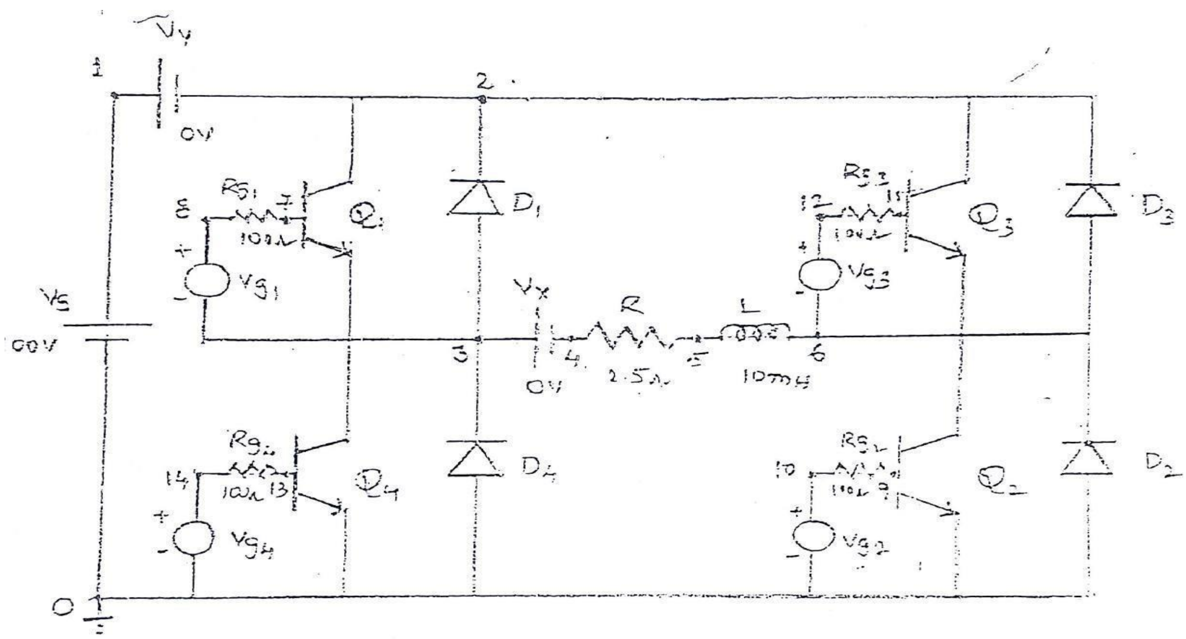


Fig - 11.1 Circuit Diagram of PSPICE Simulation of Single Phase Inverter

PROCEDURE:

1. Represent the nodes for a given circuit.
2. From desktop of your computer click on “START” menu followed by “programs” and then clicking appropriate multisim
3. Connect in the Simulator the given circuit diagram
4. To make changes in the program open the circuit file, modify, save & Run the program.

5. If there are no errors, simulation will be completed & it will be displayed with a statement as “simulation completed”.

PRE LAB QUESTIONS:

1. What is the function of inverter?
2. What is meant by carrier wave form in PWM control?
3. What are the control strategies for inverter?

POST LAB QUESTIONS:

1. What are the advantages of PWM inverter?
2. What are the applications of PWM inverter?
3. What are the advantages of simulation using PSPICE?

EXPERIMENT - 12

SINGLE PHASE DUAL CONVERTER WITH RL LOADS

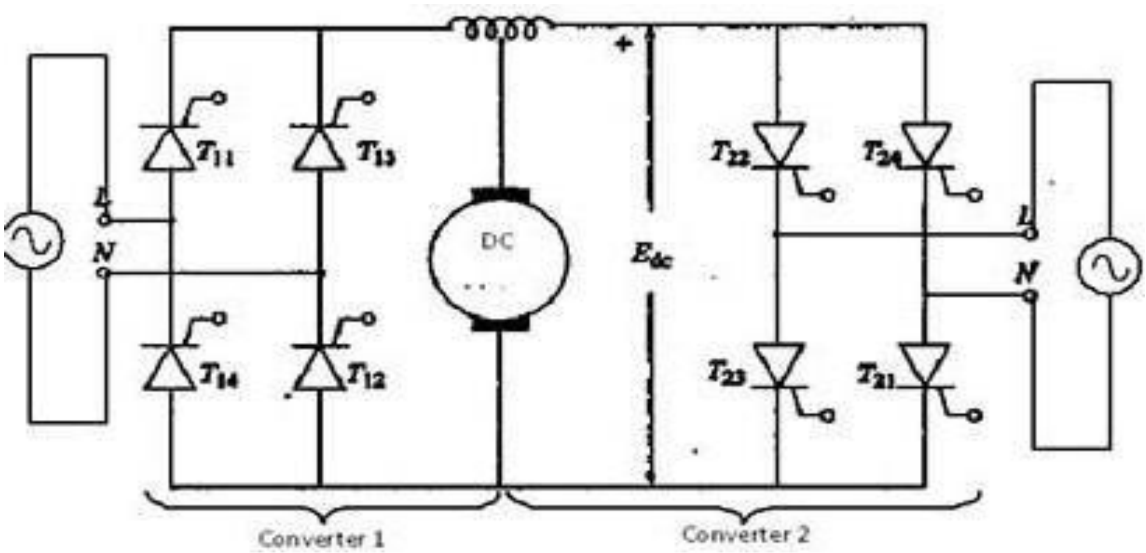
AIM:

To study the operation of single phase dual converter with RL loads.

APPARATUS:

S.No	Equipment	Range	Type	Quantity
1	Single phase dual converter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Isolation Transformer			
5	Variable Rheostat			
6	Inductor			
7	DC Voltmeter			
8	DC Ammeter			

CIRCUIT DIAGRAM:



Single Phase Dual Converter

PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Connect firstly AC supply from Isolation Transformer to circuit.

3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.
5. Connect CRO probes and observe waveforms in CRO across load and device in single phase dual converter.
6. By varying firing angle gradually up to 180^0 and observe related waveforms.
7. Measure output voltage and current by connecting AC voltmeter & Ammeter.
8. Tabulate all readings for various firing angles.
9. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
10. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
11. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

S. No	**Input Voltage (V_{in})**	**Firing angle in Degrees**	**Output voltage (V_0)**		**Output Current (I_0)**	
			Theoretical	**Practical**	**Theoretical**	**Practical**
1						
2						
3						
4						
5						
6						

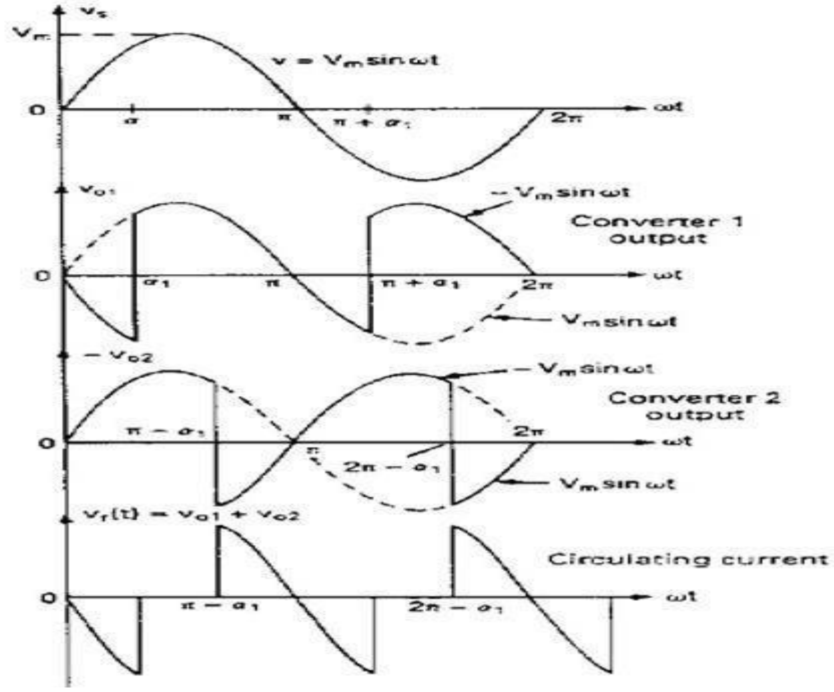
MODEL CALCULATIONS:

$V0 = (2\sqrt{2}V/\Pi) * \cos\alpha$ $I0 = (2\sqrt{2}V/\Pi R) * \cos\alpha$ α

= Firing Angle

V = RMS Value across transformer output

MODEL GRAPH:



Single Phase Dual Converter output waveforms

PRE LAB VIVA QUESTIONS:

1. What is the condition for ideal dual converter operation?
2. What are the four quadrant operations are possible with dual converter drives?
3. What is the purpose of inductor in dual converters?
4. What are the modes of operations for a dual converter?

POST LAB VIVA QUESTIONS:

1. What are the applications of dual converters?
2. Which mode of operation is suitable for four quadrant operation of dual converter?

EXPERIMENT - 13

SINGLE PHASE BRIDGE CONVERTER WITH R AND RL LOADS

Introduction

The experiment on the "Single Phase Bridge Converter with R and RL Loads" is a fundamental study in power electronics, focusing on a widely used circuit for converting alternating current (AC) to direct current (DC). A single-phase bridge converter, also known as a full-wave rectifier, utilizes four diodes arranged in a bridge configuration to efficiently convert AC input to DC output. This setup is crucial in applications ranging from household electronics to industrial power systems, where DC power is required.

The experiment involves analyzing the performance of the bridge converter under different load conditions - purely resistive (R) and a combination of resistive and inductive (RL). These load types provide insights into how the converter operates under varying electrical characteristics. The study of these configurations helps in understanding the rectification process, the impact of load on waveform characteristics, and the efficiency of the conversion process.

Learning Objectives

1. Understanding Bridge Converter Principles: Gain a comprehensive understanding of the operational principles of single-phase bridge converters, including the role of diodes in the full-wave rectification process.
2. Circuit Construction and Load Integration: Learn to construct a single-phase bridge converter circuit and integrate both resistive and resistive-inductive loads, focusing on the effects of different load types on converter performance.
3. Analysis of Output Voltage and Current Waveforms: Develop skills in analyzing the output voltage and current waveforms, understanding how load types influence the rectification efficiency, ripple factor, and waveform shape.
4. Exploring the Effect of Inductive Load: Study the influence of an inductive component in the load (RL load) on the converter's performance, particularly in terms of waveform distortion, ripple content, and the need for filtering.
5. Measurement Techniques and Instrumentation: Utilize instruments like oscilloscopes and multimeters to measure and interpret the converter's output under various load conditions.
6. Safety in Power Electronic Experiments: Emphasize the importance of safety practices when handling high-voltage AC-to-DC conversion circuits.
7. Real-World Application Scenarios: Understand the practical applications of single-phase bridge converters in various electronic and electrical systems.
8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by diagnosing and rectifying issues in the bridge converter circuit and making adjustments for optimal performance.

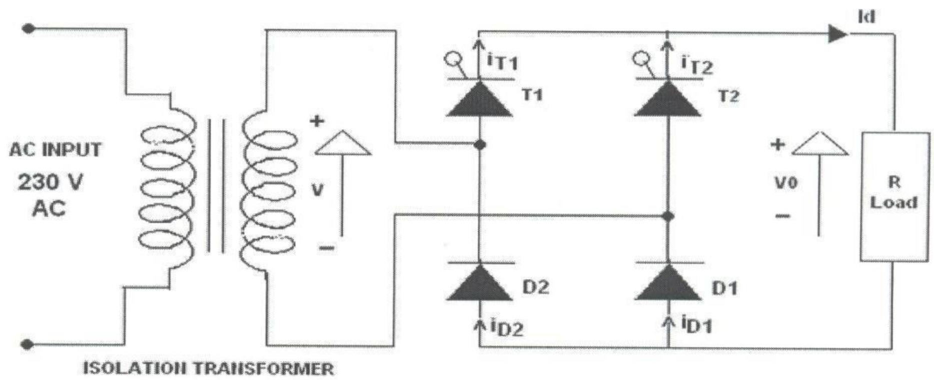
9. Data Analysis and Interpretation: Develop the ability to analyze experimental data, compare it with theoretical predictions, and understand the performance characteristics of the bridge converter under different load conditions.

By completing this experiment, students will not only gain a theoretical understanding of single-phase bridge converters but will also acquire practical skills crucial for designing and analyzing AC-to-DC conversion systems in a wide range of applications.

APPARATUS:

S. No	Equipment	Range	Type	Quantity
1	Single phase bridge converter power circuit and firing circuit			
2	CRO with deferential MODEL			
3	Patch chords and probes			
4	Isolation Transformer			
5	Variable Rheostat			
6	Inductor			
7	DC Voltmeter			
8	DC Ammeter			

CIRCUIT DIAGRAM:



Single Phase Bridge Converter

PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Connect firstly 30V AC supply from Isolation Transformer to circuit.
3. Connect firing pulses from firing circuit to Thyristors as indication in circuit.
4. Connect resistive load 200Ω / 5A to load terminals and switch ON the MCB and IRS switch and trigger output ON switch.

5. Connect CRO probes and observe waveforms in CRO across load and device in single phase bridge converter.
6. By varying firing angle gradually up to 180^0 and observe related waveforms.
7. Measure output voltage and current by connecting AC voltmeter & Ammeter.
8. Tabulate all readings for various firing angles.
9. For RL Load connect a large inductance load in series with Resistance and observe all waveforms and readings as same as above.
10. Observe the various waveforms at different points in circuit by varying the Resistive Load and Inductive Load.
11. Calculate the output voltage and current by theoretically and compare with it practically obtained values.

TABULAR COLUMN:

S. No	Input Voltage (V_{in})	Firing angle in Degrees	Output voltage (V_0)		Output Current (I_0)	
			Theoretical	Practical	Theoretical	Practical
1						
2						
3						
4						
5						

MODEL CALCULATIONS:

For R-L Load:

V_0 = $(2\sqrt{2}V/\pi) * \cos\alpha$

I_0 = $(2\sqrt{2}V/\pi R) * \cos\alpha$

α = Firing Angle

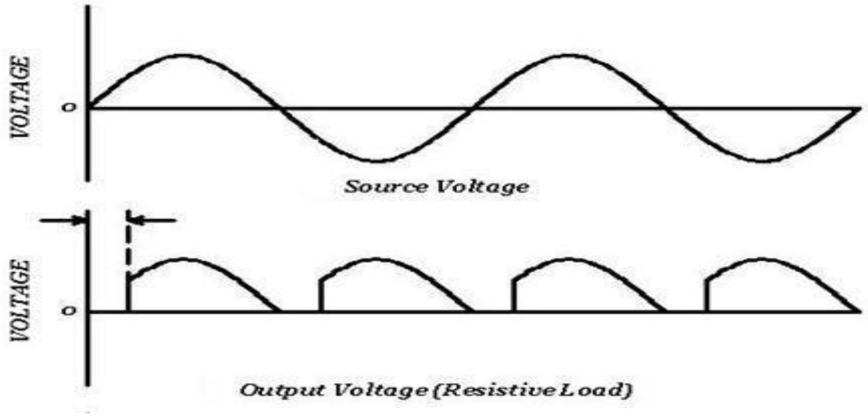
V = RMS Value across transformer output

For R Load:

$V_0 = (\sqrt{2}V/\pi) * (1+\cos\alpha)$

$I_0 = (\sqrt{2}V/\pi R) * (1+\cos\alpha)$

MODEL GRAPH:



Output waveforms of Single Phase Bridge Converter

PRE LAB VIVA QUESTIONS:

1. State the type of commutation used in this circuit.
2. What will happen if the firing angle is greater than 90 degrees?
3. What are the performance parameters of rectifier?
4. What is the difference between half wave and full wave rectifier?

POST LAB VIVA QUESTIONS:

1. If firing angle is greater than 90 degrees, the inverter circuit formed is called as?
2. What is displacement factor?
3. What is DC output voltage of single phase full wave controller?
4. What are the effects of source inductance on the output voltage of a rectifier?
5. What is commutation angle of a rectifier?
6. What are the advantages of three phase rectifier over a single phase rectifier?

EXPERIMENT – 14
GATE FIRING CIRCUITS FOR SCRS
(R - C TRIGGERING)

Introduction

The experiment on "Gate Firing Circuits for SCRs (R-C Triggering)" is a focused study in the field of power electronics, essential for understanding the control mechanisms of Silicon Controlled Rectifiers (SCRs). SCRs are widely used in applications requiring control over high power, like motor control, power regulation, and controlled rectifiers. The key to utilizing SCRs effectively lies in their gate triggering mechanism. The R-C triggering method, involving a resistor-capacitor (R-C) network, is one of the most common techniques used to provide the necessary gate pulse to turn the SCR on.

This experiment involves designing and testing an R-C gate triggering circuit for an SCR. Students will explore how changes in the R-C components affect the firing angle of the SCR, which in turn controls the conduction period and ultimately the power delivered to the load. This practical approach helps in understanding the dynamics of SCR operation and the nuances of gate pulse generation.

Learning Objectives

1. Understanding SCR and Gate Triggering Principles: Gain a thorough understanding of SCR operation, focusing on the significance of gate triggering and the role of gate current in controlling SCR conduction.

2. Design and Construction of R-C Triggering Circuit: Learn to design and construct an R-C triggering circuit for an SCR, understanding how the resistor and capacitor values determine the firing angle.

3. Analysis of Firing Angle and SCR Conduction: Develop skills in analyzing the relationship between the R-C network parameters and the SCR's firing angle, and how this influences the duration of conduction.

4. Exploring the Effects of Component Variations: Experiment with different resistor and capacitor values in the R-C network to observe how they affect the firing angle and the SCR's performance in a circuit.

5. Measurement Techniques: Use oscilloscopes and other measurement tools to observe and measure the gate-to-cathode voltage and current, and the SCR's output characteristics under different firing conditions.

6. Safety Considerations in High-Power Circuits: Emphasize the importance of safety when working with SCRs and high-power circuits, particularly when dealing with gate triggering mechanisms.

7. Real-World Application Scenarios: Understand the practical applications of SCRs in industrial and consumer electronics, and the relevance of precise gate control for effective power management.

8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by troubleshooting the R-C triggering circuit and optimizing it for desired firing angles and SCR performance.

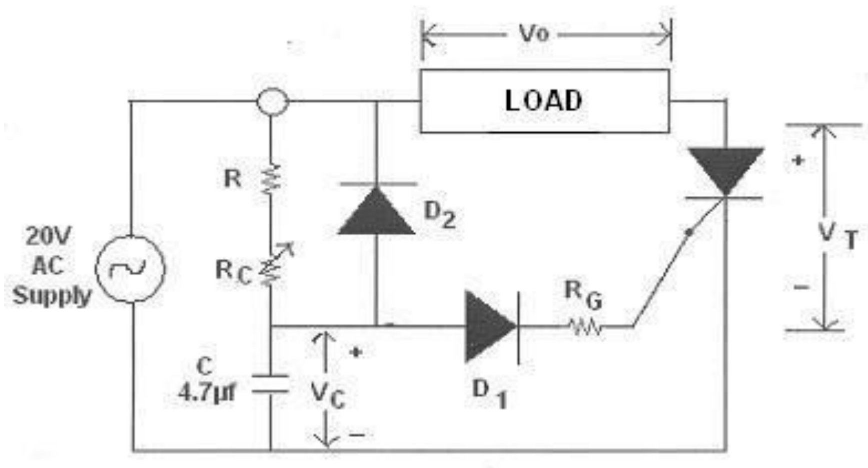
9. Data Analysis and Interpretation: Cultivate the ability to analyze experimental data, compare it with theoretical expectations, and understand the nuances of SCR gate triggering.

By completing this experiment, students will not only learn the theoretical concepts of SCR operation and R-C triggering but will also acquire practical skills essential for designing and analyzing SCR-based power control systems.

APPARATUS:

S. No	Name of the Equipment	Range	Qty
1	Resistance-Capacitance Firing Circuit		
2	Patch chords		
3	CRO with differential MODEL		
4	R-Load		

CIRCUIT DIAGRAM:



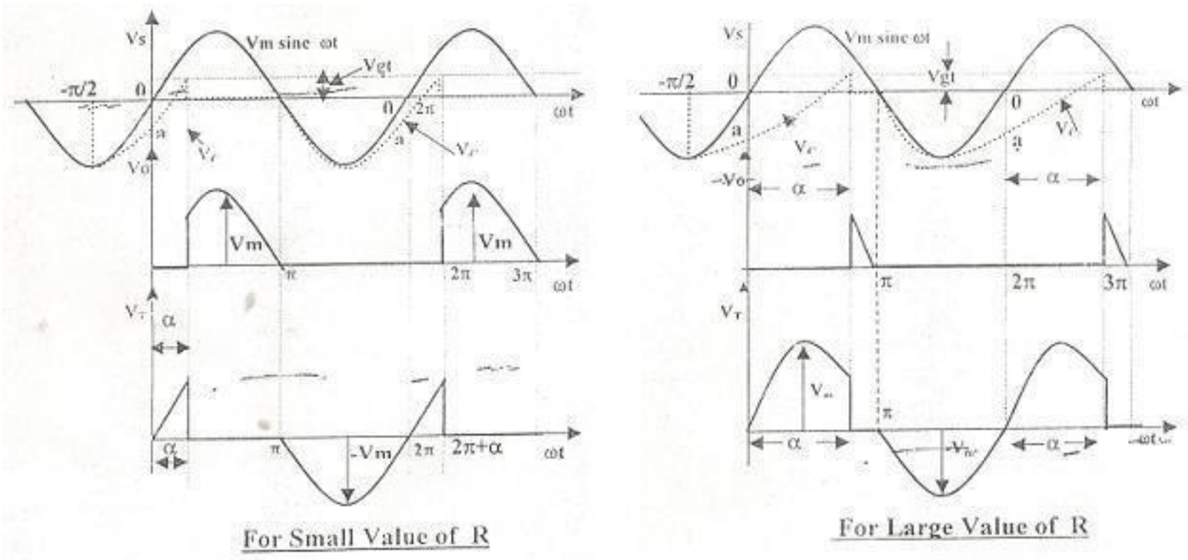
RC Triggering Circuit

PROCEDURE:

1. Make all connections as per the circuit diagram.
2. Give the AC Power supply 20V/1A from the source indicated in the front panel.
3. Connect Load i.e., Rheostat of 200Ω between two points.
4. Switch ON Power supply and observe the wave forms of input & output at a time in the CRO.[CH-1 or CH-2].
5. Slowly vary the control Resistor R_C, that firing angle can vary from 0-180°.
6. Observe various voltage waveforms across load, SCR and other points, by varying the Load Resistance and Firing R_C part.

7. Compare practical obtained voltage waveform with theoretical waveform and observe the Firing angle in R - C Triggering.

MODEL GRAPH:



Output Wave forms of RC Triggering

PRE LAB VIVA QUESTIONS:

1. Draw the characteristics of SCR.
2. Define firing angle.
3. Define extinction angle.

POST LAB VIVA QUESTIONS:

1. What are the advantages of R triggering?
2. What are the advantages of RC triggering?
3. What are the limitations of RC triggering?
4. What are the limitations of R triggering?

UJT TRIGGERING

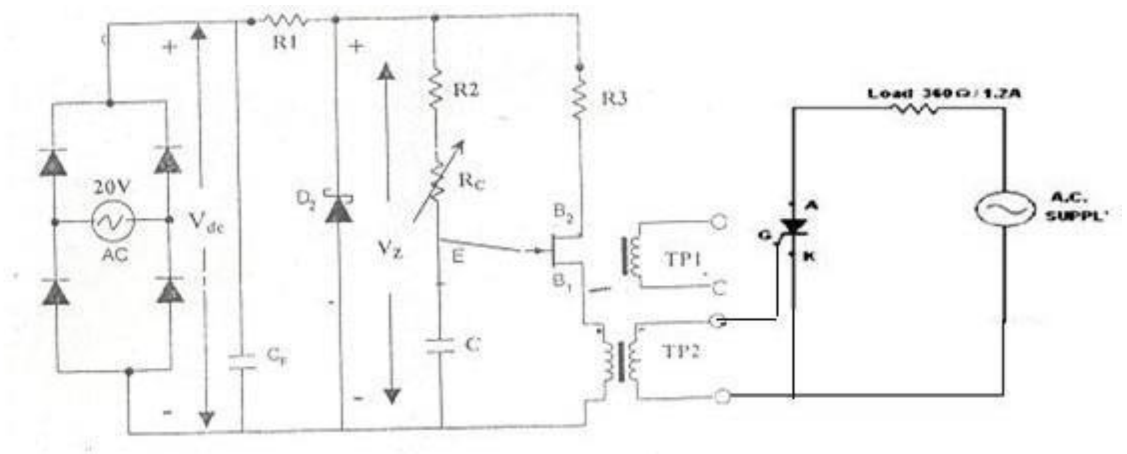
AIM:

To study Firing of SCR using UJT Relaxation Oscillator and also to study UJT Relaxation

APPARATUS:

S. No.	Name of the equipment	Range	Qty
1	UJT Firing Circuit		
2	Patch chords & Probes		
3	CRO with differential MODEL		
4	R-Load		

CIRCUIT DIAGRAM:



UJT Triggering Circuit

PROCEDURE:

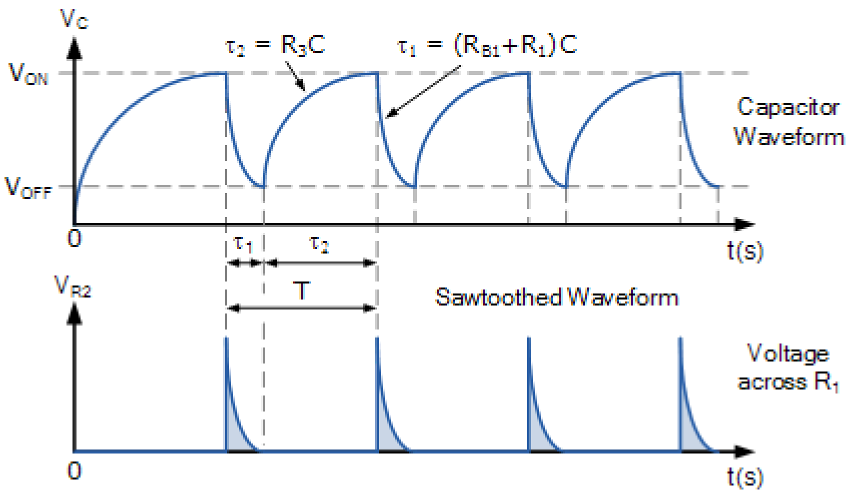
1. First observe the waveforms at different points in circuit and also trigger output T1 and T1` observe the pulses are synchronized.
2. Now make the connections as per circuit using AC source, UJT Relaxation Oscillator, SCR's and Loads.
3. Observe the waveforms across the load and SCR and other points, by varying the variable resistor RC and resistance load, observe firing angle of SCR.
4. Use differential MODEL for observing two waveforms (input and output) simultaneously in channel 1 and channel 2.

5. Check the waveforms for large value of RC and small value of RC and also triggering points of SCR.

FOR RELAXATION OSCILLATOR:

1. Short the CF capacitor to the diode bridge rectifier to get filtered AC Output.
2. We get equidistance pulses at the output of pulse transformer.
3. The frequency of pulse can be varied by varying the potentiometer.
4. Observe that capacitor charging and discharging time periods and calculate frequency and RC time constant of UJT Relaxation Oscillator by using given formulas.

MODEL GRAPH:



Output Wave Forms of UJT Triggering

PRE LAB VIVA QUESTIONS:

1. What is the ratio of latching current to holding current?
2. What is the difference between UJT triggering and temperature triggering?
3. What are the limitations of light triggering? 4. What are the advantages and disadvantages of temperature triggering?

POST LAB VIVA QUESTIONS:

1. What are the merits of UJT triggering?
2. What are the demerits of UJT triggering?

EXPERIMENT - 15

SINGLE PHASE PARALLEL INVERTER WITH R AND RL LOADS

Introduction

The experiment on the "Single Phase Parallel Inverter with R and RL Loads" is a crucial study in the field of power electronics, focusing on a specific type of inverter circuit design. Parallel inverters, also known as voltage source inverters, convert direct current (DC) into alternating current (AC) and are essential in applications like uninterruptible power supplies (UPS), renewable energy systems, and AC motor drives. In a single-phase parallel inverter, power electronic switches (such as SCRs, IGBTs, or MOSFETs) are employed in a configuration that allows the generation of an AC output from a DC input.

This experiment involves constructing a single-phase parallel inverter and analyzing its performance with resistive (R) and combined resistive-inductive (RL) loads. The study provides insights into the inverter's capability to handle different types of loads, the quality of the AC output waveform, and the efficiency of the inverter under various operating conditions.

Learning Objectives

1. Understanding Parallel Inverter Principles: Gain a comprehensive understanding of the operational principles of single-phase parallel inverters, including the role of power electronic switches in the inverter circuit.

2. Circuit Construction and Load Integration: Learn to construct a single-phase parallel inverter circuit and integrate both resistive and resistive-inductive loads, focusing on the effects of different load types on inverter performance.

3. Analysis of Output Voltage and Frequency: Develop skills in analyzing the output voltage waveform and frequency of the inverter, understanding how these parameters are influenced by the inverter design and load characteristics.

4. Exploring the Impact of Load Variations: Study how varying the load from purely resistive to resistive-inductive affects the inverter's efficiency, output voltage waveform, and harmonic content.

5. Measurement Techniques and Instrumentation: Utilize instruments like oscilloscopes and multimeters to measure and interpret the inverter's output under various load conditions.

6. Safety in Power Electronic Experiments: Emphasize the importance of safety practices when handling high-power electronic circuits, particularly those involving high-frequency switching.

7. Real-World Application Scenarios: Understand the practical applications of single-phase parallel inverters in various electronic and electrical systems.

8. Problem-Solving and Critical Thinking: Enhance problem-solving skills by diagnosing and troubleshooting issues in the inverter circuit and making adjustments for optimal performance.

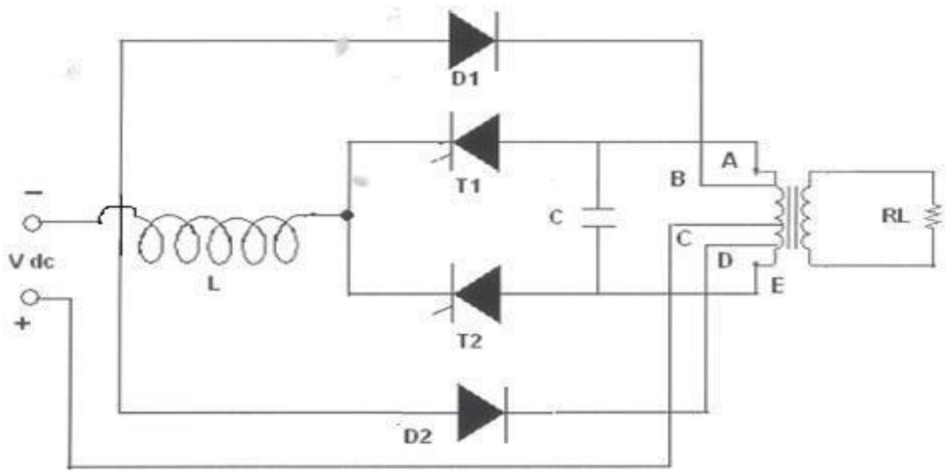
9. Data Analysis and Interpretation: Develop the ability to analyze experimental data, compare it with theoretical predictions, and draw conclusions about the performance of the parallel inverter under different load conditions.

By completing this experiment, students will not only learn the theoretical aspects of single-phase parallel inverters but will also acquire practical skills crucial for designing and analyzing power conversion systems in a wide range of applications.

APPARATUS:

S. No	Name of the Equipment	Range	Qty
1	Parallel inverter circuit.		
2	Patch chords & Probes		
3	CRO		
4	Regulated power supply		
5	R load		

CIRCUIT DIAGRAM:

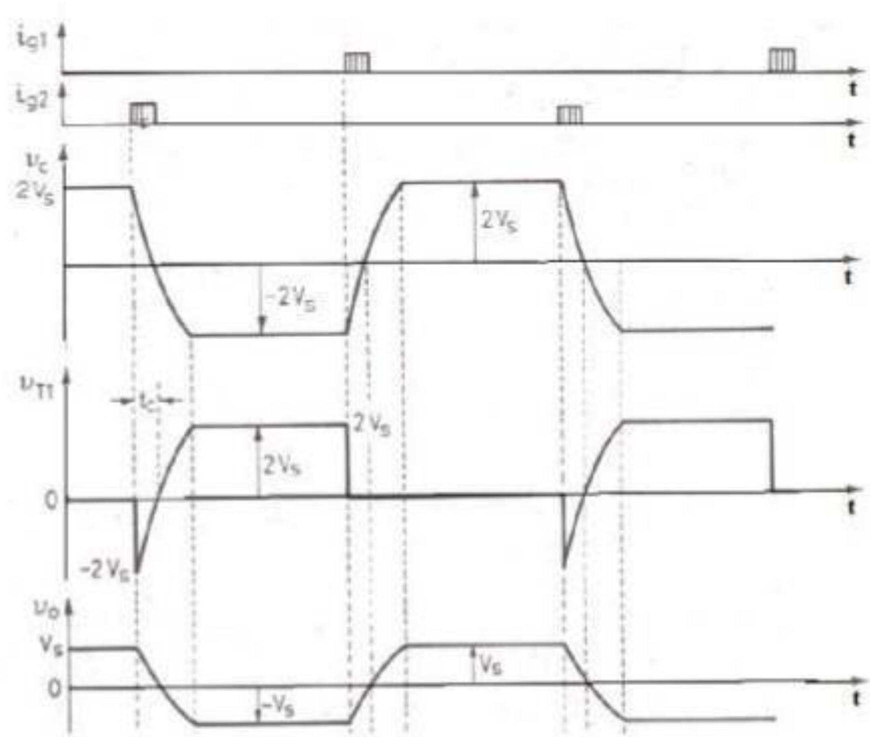


Circuit Diagram of Single Phase Parallel Inverter

PROCEDURE:

1. Make all connections as per the circuit, and give regulated power supply 30V/5A.
2. Give trigger pulses from firing circuit to gate and cathode of SCR's T1 & T2.
3. Set input voltage 15V, connect load across load terminals.
4. Now switch ON the DC supply, switch ON the trigger output pulses.
5. Observe the output voltage waveforms across load by varying the frequency pot.
6. Repeat the above same procedure for different value of L,C load values.
7. Switch off the DC supply first and then switch off the inverter. (Switch off the trigger pulses will lead to short circuit)

MODEL GRAPH:



Output Wave Forms of Single Phase Parallel Inverter

PRE LAB VIVA QUESTIONS:

1. Why is this circuit called as parallel inverter?
2. What is the type of commutation for parallel inverter?
3. What is the configuration of capacitor?

POST LAB VIVA QUESTIONS:

1. What is the dead zone of an inverter?
2. Up to what maximum voltage will the capacitor charge during circuit operation?
3. What is the amount of power delivered by capacitor?
4. What is the purpose of inductors in parallel inverters?